

A bi-level hybrid algorithm for solving multi-target inspection path planning problem of mobile robot in complex radioactive indoor environment

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ABSTRACT

Multi-target inspection path planning (MTIPP) of mobile robot (MR) represents a significant area of research in the context of environmental monitoring and routine inspection of nuclear power plants (NPPs). Given the challenges posed by complex radioactive indoor environments, characterized by the presence of numerous radioactive sources and dense obstacles, a bi-level multi-objective programming framework is proposed to model the MTIPP problem. To navigate this model effectively, a novel bi-level hybrid algorithm named ACO-GA-A* that integrates improved ant colony optimization (IACO), genetic algorithm (GA) with modified A* algorithm is developed. In the upper level, ACO with a GA-based non-uniform initial pheromone distribution, an adaptive heuristic function and an elite strategy for pheromone update is employed to determine the optimal traversal sequence of inspection targets. In the lower level, a modified A* algorithm, which considers multiple constraints including path length, risk degree and energy consumption, is utilized to plan pairwise paths between targets, thereby generating cost graphs. Comparative simulation experiments are conducted in various complexity radioactive scenarios. The results indicate that the modified A* can plan pairwise paths with lower total costs in shorter time compared to traditional A*, ACO, and GA. Furthermore, the ACO-GA-A* demonstrates better sensitivity, reliability, and convergence characteristics compared to some other bi-level hybrid algorithms. Subsequent real-world experimentation corroborates the effectiveness and feasibility of both the bi-level programming framework for MTIPP and the proposed ACO-GA-A* algorithm.

1. Introduction

Radiation brings a great damage to radiological occupational personnel (POP) working in nuclear power plants (NPPs). Specifically, when the cumulative dose received by the human body increased by 1 Gy, the risk of death from solid tumors increased by 52 % (David et al., 2023). Therefore, in recent years, replacing the manual works by intelligent equipment is an essential demand for the digital, intelligent and safe development of NPPs. Up to now, many radiation-protected mobile robots (MRs) mounted with sensors (e.g., radiation detector, LiDAR, camera etc.) developed for nuclear-related activities (e.g., nuclear decommissioning (Bird et al., 2021), radiation monitoring (Ducros et al., 2017) and nuclear accident emergency (Nagatani et al., 2013)) have dual benefits, it extend the repetitive and difficult tasks that cannot

be performed manually in hazardous areas and minimize the risk of radiation for POP in NPPs. In this context, more and more efforts have been devoted to the researches of these MRs focusing on sectors such as radiation shielding, radiation field reconstruction, autonomous navigation, remote control and communication. In particular, the path planning (PP), as a hot topic for MRs autonomous navigations (Adibeli et al., 2021; Chen, Wu, et al., 2022; Zhang, Luo, et al., 2023; Zhang, Yin, et al., 2023), provides an effective means to improve the efficiency, stability and safety of MRs in performing special tasks in this extremely hazardous environment.

To meet the challenges of PP of MRs in radioactive environments, many research efforts have been devoted as detailed in Section 2.1. However, most existing works (Huang et al., 2022; Li et al., 2016; Liu et al., 2015; Wei et al., 2013; Zhang, Yin, et al., 2023) paid attention to

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point-to-point path planning (PTPPP) problem, which focus on finding an optimal path with minimum cost from a start point to a destination point. Only a few works considered the specific mission requirements. Thus, in this paper, we focus on a common class of target traveling missions, e.g., nuclear facility routine inspection and radiation monitoring, which have the same requirement to travels a set of specific targets once and only once. The PP of such missions is called multi-target inspection path planning (MTIPP), which likes an N-P hard traveling salesman problem (TSP). Moreover, cost graph of between each pair of inspection targets should be built to evaluate the traveling order determination. To obtain the cost graphs, it is necessary to perform PTPPP between each pair of targets by considering multiple constraints, e.g., path length, radiation risk and energy consumption, etc.

Furthermore, in the literature, a large amount of hybrid algorithms have been developed for solving the MTIPP problems in conventional environments, as discussed in Section 2.2. However, there are few PP approaches specially designed for the MTIPP problem constrained by multiple factors, similar to the multi-objective traveling salesman problem (MOTSP) (Khan et al., 2023; Khanra et al., 2019; Yu et al., 2022). Nevertheless, such constraints must be considered when MRs are arranged in radioactive environments to conduct the target traveling missions due to the limited radiation tolerance and power supply. In addition, using conventional iteration-based intelligent algorithms (e.g., ant colony optimization (ACO) and genetic algorithm (GA)) to plan pairwise path for obtaining cost graphs between each targets will lead to a great time loss (Sui et al., 2023; Yu et al., 2019).

Inspired by the above analysis, in this paper, the MTIPP problem of target traveling missions in complex radioactive indoor environment is modeled by a bi-level programming framework and solved by a bi-level hybrid algorithm. The main contributions of this paper are as follows:

- 1) A bi-level programming framework for MTIPP of MR is constructed. The upper level is a MOTSP to yield sequence of inspection targets based on cost matrices calculated by the lower level. The lower level is a pairwise PP based on two-layer cost environmental model aim at minimizing the path length, risk degree (cumulative radiation dose) and turning energy consumption.
- 2) A novel bi-level hybrid algorithm, designated ACO-GA-A*, is developed to solve the bi-level model of MTIPP. In the upper level, ACO with GA-based non-uniform initial pheromone distribution, adaptive heuristic function and elite strategy for pheromone update is utilized to optimize traversal sequence of inspection targets. In the lower-level, a modified A*, which considers multiple constraints into evaluation function and accommodates high-risk area avoidance strategy, is deployed to plan pairwise path between each targets for obtaining cost matrices. The optimal inspection path of MR is obtained through the information interaction of each layer continuously until the optimal solution converges.
- 3) The designed multilevel simulation experiments are carried out to validate the superiority of the proposed bi-level hybrid algorithm ACO-GA-A* compare with other algorithms in solving MTIPP problems. Real-world experiment is conducted to corroborate the effectiveness and feasibility of the MTIPP approach.

The remainder of this paper is organized as follows. Section 2 reviews the related works. Section 3 describes the MTIPP problem of MR in radioactive environment. Section 4 presents the environmental model and mathematical model of the MTIPP problem. Section 5 introduces the proposed bi-level hybrid algorithm. Section 6 discusses the experiments and result analysis. Section 7 draws conclusions and outlines potential future works.

2. Literature review

The MTIPP problem of MRs in a radioactive environment can be considered as a TSP constrained with multiple factors, such as path

length, risk and energy consumption. Hence, the relevant works are traced under two streams: the basic work of PP approaches in radioactive environment, algorithms for MTIPP, especially hybrid optimization algorithms.

2.1. PP approaches in radioactive environment

Existing PP approaches can be classified into two types, PTPPP algorithm and MTIPP algorithm, according the mission requirements of MRs in radioactive environment. As for PTPPP problems, there are popular deterministic algorithms, such as Dijkstra algorithm (Chizhov et al., 2017; Pei et al., 2020), A* algorithm (Huang et al., 2022; Li et al., 2016; Liu et al., 2015) and its variants (Chen et al., 2020; Tao et al., 2018; Tian et al., 2023), RRT* (Chao et al., 2017) and its improved versions (Chao et al., 2018; Chao et al., 2019) have been proposed. These algorithms have been verified effectively to find the optimal path with minimum radiation dose for satisfying the principle of as low as reasonably achievable (ALARA). These algorithms tend to be time-consuming while handling with PP problems in large complex environments. And, some non-deterministic algorithms, such as ACO (Wei et al., 2013), particle swarm optimization (PSO) (Chen, Wu, et al., 2022), GA (Lai, & Smith, 2022) and beetle antennae search (BAS) algorithm (Tan et al., 2020) have been applied due to their ability to dealing with multi-objective optimization problems. In addition, to enhance the efficiency and quality of single non-deterministic algorithms, some hybrid algorithms, such as PSO-APF (Li, Wei, et al., 2020), IACO-A* (Zhang, Luo, et al., 2023), IACO-A*-PSO (Zhang, Yin, et al., 2023) have been developed.

In real-world scenarios, such as facility inspection (Hong et al., 2023), radiation monitoring (Abd Rahman et al., 2022; Woller, & Kulich, 2022) and nuclear accident emergency (Tang et al., 2022), MTIPP problems are more prevalent than PTPPP. Therefore, the mathematical model of PTPPP and corresponding algorithms tend to become incapable since they are only designed for pairwise PP. Although some intelligent algorithms (Liu et al., 2014) and its hybrid versions (e.g., PSO with chaos optimization algorithm (PSO-COA) (Wang, & Cai, 2018), ACO-COA (Xie et al., 2022) and Rao-combined artificial bee colony (ABC) algorithm (Xie et al., 2022)) have been developed to solve the MTIPP problems. However, these algorithms focus on generating a path that travels all targets with a minimum radiation cost for satisfying the principle of ALARA. Thus, there is still a great challenge to construct reasonable mathematical model and programming framework for describing MTIPP problem constrained with multiple factors and develop effective algorithms for solving these problems.

2.2. Hybrid optimization algorithms for MTIPP

The MTIPP problem traversing multiple targets has been studied by a large number of scholars. In recent years, intelligent optimization algorithms such as ACO, GA, PSO and neural network (NN) have shown their significant advantages to solve TSPs compared with precise algorithm (e.g., branch and bound, dynamic programming and integer linear programming etc.) and local search algorithms including 2-Opt and 3-Opt algorithm (Zhang et al., 2022). However, limited by the principle of iteration optimization, the single intelligent algorithms still exhibits low accuracy, slow convergence, vulnerability to fall into a local optimum and high complexity of parameters setting, which will weaken its ability to solve complex TSPs. To this end, many hybrid algorithms (Deng et al., 2012; Deng et al., 2022) combining two or more algorithms have been developed and achieved great success in solving TSPs. As the most popular population-based intelligent optimization algorithms, ACO and GA with strong parallelism and global search ability have been integrated with other algorithms to enhance the performance of solving TSPs in various ways (Toaza, & Esztergár-Kiss, 2023). For ACO-based hybrid algorithm, literature (Deng et al., 2012) presented a novel hybrid algorithm GA-PSO-ACO, which optimize the initial pheromone

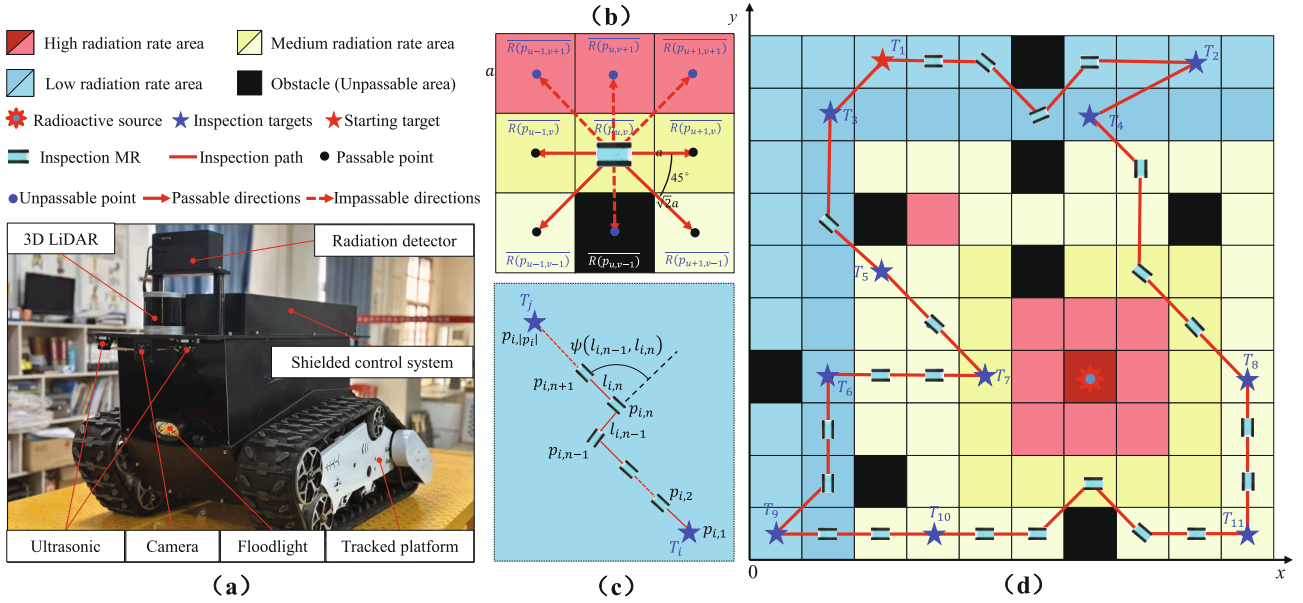


Fig. 1. The MTIPP problem for MRs in radioactive environment. (a) the structure composition of the inspection MR designed by our team; (b) search strategy based on eight directions; (c) pairwise path between two inspection target; (d) schematic diagram of planned inspection path.

allocation of ACO by GA and PSO to improve the quality of its solution to TSP. Paper (Gulcu et al., 2018) combined the ACO and 3-Opt algorithm to avoid local minima and improve the efficient and reliable of TSP. In literature (Wang, & Han, 2021), symbiotic organisms search (SOS) is used to adaptively optimize the parameters of ACO to improve the ability to solve TSP. Paper (Zhou et al., 2022) designed a parameter adaptation-based ant colony optimization algorithm based on PSO with the global optimization ability, fuzzy system with the fuzzy reasoning ability and 3-Opt algorithm with local search ability to improve the optimization ability of TSP. For GA-based hybrid algorithm, literature (Alipour et al., 2018) incorporated GA with multi-agent reinforcement learning (MARL) to obtain a better TSP solution in a shorter time. Paper (Ha et al., 2020) proposed a hybrid genetic algorithm with dynamic population management and adaptive diversity control based on a split algorithm to solve the traveling salesman problem with drone. In literature (Chen, Hu, et al., 2022), the planning results by chaotic neural network (CNN) are encoded into individuals to replace the individuals with the worst fitness in the population of GA, which has been proved to perform better than other algorithms in the inspection PP of robot in wind farm.

Indeed, while hybrid algorithms mentioned above have shown promise in proving improved solutions for TSPs, there is a notable gap in research focusing on solving MTIPP problems, similar to MOTSP (Khan et al., 2023; Khanra et al., 2019; Yu et al., 2022), in which multiple constraints should be considered. In addition, most existing literatures on TSP implicitly assumes fixed traveling costs between targets, which are crisp in nature (Changdar et al., 2014). Therefore, hybrid algorithms based on iterative mechanisms will result in significant time cost when dealing with TSPs featuring obstacles and unknown costs between targets.

3. Problem statement

3.1. MTIPP problem description

In this study, we focus on MTIPP problem of MRs, caused by multi-target inspection missions, such as nuclear facility routine inspection and radiation monitoring, with the requirement to traverse a class of predetermined inspection targets in a static radioactive indoor environment with known obstacles and radioactive sources. The problem is

displayed in Fig. 1. To describe the special problem accurately, it is illustrated from the following aspects:

Inspection MR: to obtain prior environmental information accurately, the MR are typically mounted with multi-sensor perception system consists with sensors (e.g., LiDAR, camera and ultrasonic) for sensing geometric information and radiation sensor (e.g., Geiger-Miller tubes, scintillator detectors, semiconductor detectors, etc.) for sensing radiation information. Fig. 1 (a) shows the structure composition of inspection MR. MRs usually follow a path covering predefined inspection targets, due to the limitation communication that inhibited by the reactor buildings contained thick concrete wall.

Radioactive environment: the problem considers the complex radioactive environments with high risky area with high radiation rate and dense obstacles (e.g., nuclear facilities, obstacles and walls). During mission execution, MRs should avoid high risky areas and have no collision with obstacles. In a complex environment with dense radiation sources and obstacles, two random inspection targets may be disconnected and there are impassable paths between them. For ease of description, the passability and connectivity of points and paths are defined as definitions 1 and 2.

Definition 1. ((passable & passability)) A point can be defined as a *passable point* if it is in a non-high radiation rate area (i.e., the medium radiation rate area colored in yellow or the low radiation rate area colored in blue) and there is no collision with any obstacles. Otherwise, it is an impassable point. As can be seen in Fig. 1 (b), the three points above the robot (colored red, high radiation rate area) and the point directly below the robot (colored black, obstacles) are impassable points. A *passable path* means that the points through are all passable points. A *passable region* is a collection of passable points. The directions in which the current position of the robot points to the passable points are regarded as the *passable directions*.

Definition 2. ((connected & connectivity)) Two passable points are said to be *connected* if there is a feasible path between them. And, the *connected area* means that any points in the passable area are connected.

MR inspective target traveling: the traveling mission comprising a set of inspection targets ($T = \{T_1, T_2, \dots, T_m\}$) that the MR will traverse. In other words, the MR need to travel all the inspection targets one by one from the starting target (T_1), and finally return to T_1 . The MR path planner focuses on finding an optimal passable path with minimum cost

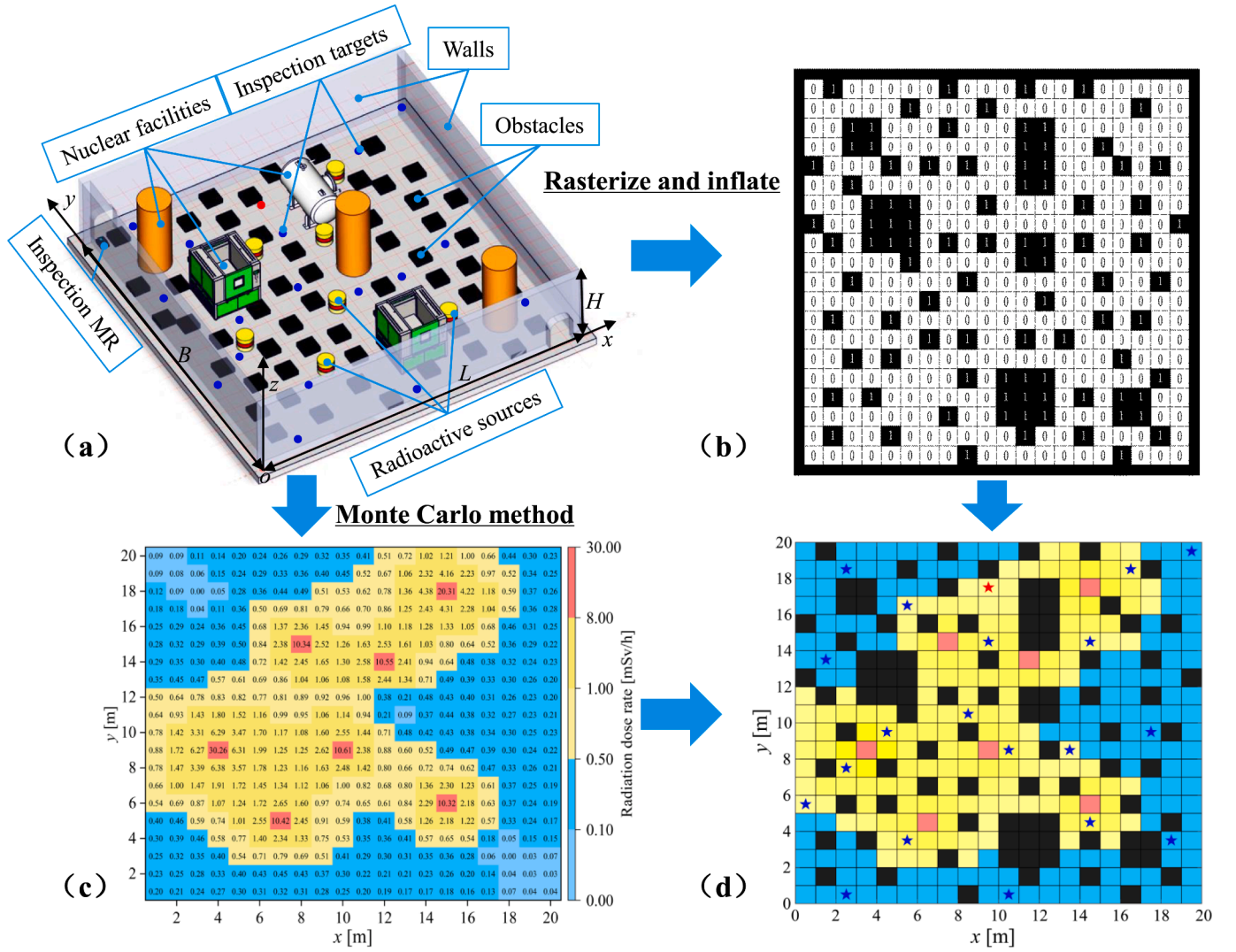


Fig. 2. The modelling process of the radioactive indoor environment. (a) 3D model of hypothetical NPP; (b) navigation layer; (c) radiation layer (risk layer); (d) two-layer cost grid map.

to traverse these targets in a certain order.

MR inspection path: for generating MR pairwise paths (i.e., optimal paths between two inspection targets), as depicted in Fig. 1 (c), the straight line method (Kanayama, & Yuta, 1988), a simple and popular existing methodology, is employed in this paper. To generate a pairwise path between two targets, an ordered set of intermediate waypoints is located so that each straight path connecting these waypoints is passable. The pairwise path is subsequently constructed from these straight segments. Consequently, the MR inspection path is a polygonal path line formed by the pairwise paths selected based on the order of inspection target traversal, as illustrated in Fig. 1 (d).

3.2. Inspection path optimize objective

To plan inspection paths in complex indoor environments featuring radioactive sources and obstacles, three essential requirements should be considered into the optimize objective, i.e., obstacle avoidance, high-risk area avoidance, and cost minimization.

Obstacle avoidance. The obstacle avoidance is a hard constraint, meaning that the MR path collides with any obstacle within the indoor environment, the planned path is deemed to be unpassable.

High-risk area avoidance. In this study, the radiation risky area characterize by distance-attenuation properties is considered, which are

prevalent in NPPs. Once the MR enters such an area, it becomes endangered, with the risk increasing as the MR approaches the center of the radioactive source. Radiation risky areas are generated by gamma point sources (Li, Wei, et al., 2020). Considering the maximal risk intensity (i.e., air-absorbed dose rate) as RI , the risk intensity of the random point p is inversely proportional to the square of its distance r to the center of the radioactive source. In actual inspection tasks, the MR will be failure due to the damage to its sensitive components when the risk intensity exceeds the tolerable range. Assuming the maximal risk intensity that the MR can tolerate is RI_{max} , the risk intensity RI_r at all passable points along the planned inspection path should below RI_{max} to avoid high risk areas. This relationship can be illustrated as shown in Eq. (1).

$$RI_r = \frac{RI\Gamma_k}{r^2} < RI_{max} \quad (1)$$

where, Γ_k is the air kerma rate constant.

Cost minimization. Referring to existing works such as (Zhang, Luo, et al., 2023; Zhang, Yin, et al., 2023), the MR inspection cost considers path length, risk degree, and energy consumption. The risk degree cost, also denoted as cumulative radiation dose cost, is defined by the total radiation dose received by MR along the planned path. The energy consumption cost is calculated by assessing the angles between adjacent

Table 1

Explanation of notations and equation terms.

Mathematical model		Proposed algorithm	
Terms	Definitions and descriptions	Terms	Definitions and descriptions
T	A set of all the inspection targets.	$\tau_k^t(i, j)$	The pheromone between target T_i and T_j in the t th iteration.
m	Total number of inspection targets.	$\eta_k(i, j)$	Heuristic information.
T_i	Inspection target i .	α	pheromone concentration factor.
e_{ij}^q	The q th parametric value (cost value) between target T_i and T_j	β	expected heuristic factor.
$\mathcal{F}_q$	The total cost of the inspection path.	J_k	The set of inspection targets that ant k has not traveled.
ω_q	Cost weight coefficients that can be used to adjust the preference for different costs.	q_0	Constant parameter.
I	A set for recording the traversal sequence of all inspection targets.	q	Random parameter in [0, 1].
s_i	The i th inspection target in the I .	$P_k^t(i, j)$	Transition probability from T_i to T_j for the k th ant in the t th iteration.
$e_{s_i, s_{i+1}}^q$	The q th parametric value between target T_i and T_{i+1} .	ξ	Adaptive adjustment coefficient of ACO.
p_i	The set of waypoints between target T_i and T_j .	I_{max}	The maximum iteration of ACO.
$M(p_i)$	The total cost between the target T_i and T_j .	I_c	Current iteration of ACO.
$L(p_i)$	The path length cost between the target T_i and T_j .	$\tau^t(i, j)$	The pheromones on the path at the iteration t .
$R(p_i)$	The risk cost between the target T_i and T_j .	ρ	Pheromone volatilization factor.
$E(p_i)$	The turning energy consumption cost between the target T_i and T_j .	Q	Constant parameter for indicating the pheromone increment coefficient.
$d(p_{i,n}, p_{i,n+1})$	Path length from the waypoint $p_{i,n}$ to $p_{i,n+1}$. (m)	$\Delta\tau_k(i, j)$	Pheromone concentration released by the ant k on the path (i, j) between T_j and T_j .
$(x_{i,n}, y_{i,n})$	The coordinate value of the center point of the grid where the waypoint $p_{i,n}$ is located.	N_{max}	The maximum number of ants.
$R(p_{i,n}, p_{i,n+1})$	Cumulative radiation dose from the waypoint $p_{i,n}$ to $p_{i,n+1}$. (mSv)	G_{max}	The maximum generation of GA.
$\overline{R(p_{i,n})}$	Average radiation dose rate in the grid $p_{i,n}$. (mSv/h)	n_p	The number of population of GA.
a	Length of each grid. (m)	p_c	Crossover probability of GA.
v	Velocity of the MR. (m/s)	p_m	Variation probability of GA.
$\psi(l_{i,n-1}, l_{i,n})$	The angle variable from the line $l_{i,n-1}$ that connects grid $p_{i,n-1}$ and $p_{i,n}$ to the line $l_{i,n}$ that connects grid $p_{i,n}$ and $p_{i,n+1}$. (°)	gap	Generation gap of GA.
N_{turn}	The sum of turns between the target T_i to T_j .	$f(n)$	Total cost of planned path.
RI_{max}	The maximum radiation dose rate that the MR can tolerate.	$g(n)$	Cost of planned path from starting point to current point.
		$h(n)$	Cost of a planned path from current point to goal point.
		$g(p_{i,x})$	The cost from starting waypoint to the current waypoint $p_{i,x}$ for modified A* algorithm.
		$h(p_{i,x})$	The cost from current waypoint $p_{i,x}$ to the goal waypoint for modified A* algorithm.

path segments (Miao et al., 2021).

To address the multi-objective path planning (MOPP) problem, two approaches have been developed as follows: 1) specialize multi-objective optimization algorithms are employed to obtain Pareto optimum solutions; 2) weight summation method (WSM) is utilized to transform the multi-objective optimization problem into a single optimization problem, which is then optimized using specialized algorithms. Among these approaches, WSM is widely adopted due to its popularity and ease of use (Sui et al., 2023; Yu et al., 2019). Moreover, WSM allows for the generation of ideal paths tailored to specific missions and user preferences, such as shortest path, safest path, and minimum energy path, among others. Thus, in this study, WSM is applied to calculate the total cost of an inspection path. Specially, three cost weight coefficients, denoted as ω_1 , ω_2 , ω_3 are adopted for path length cost, risk degree cost and energy consumption cost, respectively.

Mathematically, given a path from start point S to destination point D with n waypoints, i.e., $Path(S, D) = (s_1, s_2, \dots, s_i, s_{i+1}, \dots, s_{n-1}, s_n)$ with $S = s_1$ and $D = s_n$, the total cost $\mathcal{F}(S, D)$ of the inspection path can be calculated by the Eq.(2).

$$\mathcal{F}(S, D) = \omega_1 \sum_{i=1}^{i=n} L(s_i, s_{i+1}) + \omega_2 \sum_{i=1}^{i=n} R(s_i, s_{i+1}) + \omega_3 \sum_{i=2}^{i=n-1} E(s_{i-1}, s_i, s_{i+1}) \quad (2)$$

where, $L(s_i, s_{i+1})$ gets the path length cost from s_i to s_{i+1} ; $R(s_i, s_{i+1})$ represents the risk degree cost of the path segment from s_i to s_{i+1} ; $E(s_{i-1}, s_i, s_{i+1})$ denotes the turning energy consumption between the two path segments that joint at s_i ; The detailed calculation methods will be introduced in Section 4.3.

3.3. Problem assumptions

To facilitate the mathematical modeling of MTIPP, some assumptions are illustrated in this paper as follows:

- 1) The MR is regarded as a particle due to its negligible size compared to the environment. It can move along eight possible directions in the grid map at a constant velocity, as depicted in Fig. 1 (b);
- 2) Before the PP procedure runs, the obstacle information of the environment has been detected by the 3D LiDAR mounted on the MR;
- 3) The position of radiation sources is fixed, and their intensity does not decay over time;
- 4) Each inspection target is treated as a passable point (grid), and its position remains unchanged;
- 5) Sufficient energy is provided for the MR to traverse all the inspection targets.

4. Models for MTIPP problem

In this section, a novel environmental modelling method is proposed to describe indoor environments with radioactive sources and obstacles, along with a mathematical model expressing the MTIPP problem of inspecting target travel missions.

4.1. Radioactive indoor environment model

With the confidentiality of NPPs, we create a hypothetical environment model of a building based on the physical characteristics present in real NPP environments. To simulate the realistic radioactive indoor environment, the two-layer cost grid map is utilized, which combines a radiation layer based on radiation information with a navigation layer based on obstacle information. With this map model, accurate calculation of the length and cumulative radiation dose of the planned path is achievable. The modelling process is illustrated in Fig. 2. Furthermore, the construction methods of the navigation layer and radiation layer are detailed as follows:

Navigation layer: Given that the 3D model of the NPP is a cube of

dimensions $L \times B \times H$, containing obstacles such as nuclear facilities (e.g., containers, control boxes, walls) and other convex objects that hinder the movement of MRs, the model is projected on the xoy plane. Subsequently, the plane is rasterized to mark the locations of the obstacles, and the obstacles are inflated to reduce the risk of collision between the MR and obstacles. In addition, the number '1' is used to represent the binary grid that its height exceeds the obstacle-surmounting ability of the MR, and the number '0' represents passable areas. The binary grid representation is illustrated in Fig. 2 (b).

Radiation layer (risk layer): The location and intensity of radioactive sources are obtained by the inspection MR in this paper. Subsequently, the Monte Carlo Code (Xiao, & Cai, 2022), which is more suitable for complex scenes than the point kernel method, is adopted to calculate the radiation dose rate value of each grid. It involves defining the physical properties of the structures and considering the radiation shielding provided by nuclear facilities. The visualization of the radiation map is depicted in Fig. 2 (c), where red, yellow and blue represent high, medium and low radiation dose rates, respectively.

4.2. Definition of equation terms used in this paper

In the development of mathematical model and the proposed algorithm, various notations and equation terms are used, which are categorized and explained in Table 1.

4.3. Mathematical model of MTIPP problem

In this section, a bi-level programming framework is introduced to mathematically describe the MTIPP problem of MRs inspecting in an indoor environment under the influence of multiple constraints. The upper-level programming is a MOTSP, aiming to optimizing the traversal sequence of inspection targets. This is achieved based on cost graphs (i.e., cost matrices) calculated by the lower-level programming. The lower-level is a pairwise MOPP problem, focus on finding optimal pairwise paths with minimum total cost between each pair of inspection targets. It is worth noting that the bi-level programming is different from multi-objective optimization. The objectives of each level in bi-level programming can be same or not in conflict, while the objectives of multi-objective optimization are mutually conflicting (Sui et al., 2023).

The upper-level programming: Assume that the inspection area of a NPP Ω is a rectangle of $L \times B$ on the xoy plane, containing several radioactive sources and obstacles, as shown as Fig. 2 (a). Consider the MOTSP consists of a set of m inspection targets $T = \{T_1, T_2, \dots, T_m\}$, and a two-dimensional cost matrix $(e_{ij}^q)_{m \times m}$, where $(e_{ij}^q)_{m \times m}$ represents q parametric values between T_i and T_j . Therefore, the goal of the MOSTP is to establish a Hamiltonian circuit (HC) (Khan et al., 2023) over the targets of T that optimizes path length, risk and energy consumption simultaneously. Hence, the upper-level programming model can be mathematically formulated as:

$$\begin{cases} \text{Min. } \mathcal{T}_q = \sum_{i \in T} \sum_{j \in T} (\tau_{ij} \omega_q e_{ij}^q), & q = 1, 2, 3 \\ \text{s.t. } \tau_{ij} = \begin{cases} 1, & \text{if the MR travels from } T_i \text{ to } T_j \\ 0, & \text{else} \end{cases} \end{cases} \quad (3)$$

where, e_{ij}^q denotes q th parametric value between T_i and T_j . In this paper, e_{ij}^1 , e_{ij}^2 and e_{ij}^3 represent the path length cost matrix e_{ij}^L , risk cost matrix e_{ij}^R and turning energy consumption cost matrix e_{ij}^E , respectively, which obtained by the lower-level programming described later.

Let $I = \{s_1, s_2, \dots, s_m, s_1\}$ as an inspection path of MR travels all targets and back to the start target, where $(s_i \in T) \cap (s_i \neq s_j), \forall i \in \{1, 2, \dots, m\}$. The goal of the model is to find a set I_d ($d \in \{1, 2, \dots, m!\}$) to minimize the value of the function $\mathcal{T}_q$:

$$\mathcal{T}_q = \sum_{i=1}^{m-1} e_{s_i, s_{i+1}}^q + e_{s_m, s_1}^q, q = 1, 2, 3 \quad (4)$$

The lower-level programming: This programming is designed to provide the parametric values (i.e., cost matrices) for the upper-level programming. Assume the waypoints between the inspection targets T_i and T_j as $p_i = \{p_{i,1}, p_{i,2}, \dots, p_{i,n-1}, p_{i,n}, p_{i,n+1}, \dots, p_{i,|p_i|}\}$, where $|p_i|$ is the number of waypoints between the two targets, $p_{i,1} = T_i$ and $p_{i,|p_i|} = T_j$, as shown in Fig. 1 (c). The pairwise path between each target is constrained by path length $L(p_i)$, risk degree $R(p_i)$ and turning energy consumption $E(p_i)$. Using the WSM, path evaluation function for calculating cost matrices can be defined in Eq. (5). The smaller $M(T_i \rightarrow T_j)$, the better the path.

$$M(T_i \rightarrow T_j) = M(p_i) = \omega_1 L(p_i) + \omega_2 R(p_i) + \omega_3 E(p_i) \quad (5)$$

where, $M(p_i)$ is the total cost of the path between target T_i and T_j . The three constrained indexes can be calculated as follows:

Path length: For the path length $L(p_i)$ from T_i to T_j , we calculate the sum of Euclidean Distances for all neighboring waypoints of the planned pairwise path, as shown in Eq. (6).

$$L(p_i) = \sum_{n=1}^{|p_i|-1} d(p_{i,n}, p_{i,n+1}) = \sum_{n=1}^{|p_i|-1} \sqrt{(x_{i,n+1} - x_{i,n})^2 + (y_{i,n+1} - y_{i,n})^2} \quad (6)$$

where, $d(p_{i,n}, p_{i,n+1})$ represents the Euclidean Distances between waypoints $p_{i,n}$ and $p_{i,n+1}$. In a grid-based map, the distance $d(p_{i,n}, p_{i,n+1})$ equals $\sqrt{2}$ while the MR travels along diagonal line; otherwise, the distance equals 1.

Risk degree: The risk degree, represented by the cumulative radiation dose $R(p_i)$, indicates the total radiation exposure encountered by the MR. In a discrete grid map, the $R(p_i)$ from target T_i to T_j can be calculated by the following formula:

$$\begin{aligned} R(p_i) &= \sum_{n=1}^{|p_i|-1} R(p_{i,n}, p_{i,n+1}) \\ &= \sum_{n=1}^{|p_i|-1} d(p_{i,n}, p_{i,n+1}) \times \frac{\overline{R(p_{i,n})} + \overline{R(p_{i,n+1})}}{2} \times \frac{a}{v} \end{aligned} \quad (7)$$

where, $R(p_{i,n}, p_{i,n+1})$ signifies the cumulative radiation dose between waypoints $p_{i,n}$ and $p_{i,n+1}$; $\overline{R(p_{i,n})}$ denotes the average radiation dose rate within the grid $p_{i,n}$; a is the length of each grid; v is the velocity of the MR.

Turning energy consumption: The MR consumes more energy when changing course, thus minimizing the number of turns and turning angles is preferable for achieving a smoother path (Miao et al., 2021). Turning energy consumption can be calculated by the Eq. (8) and (9).

$$E(p_i) = \varepsilon_1 \sum_{n=2}^{|p_i|-1} \psi(l_{i,n-1}, l_{i,n}) + \varepsilon_2 N_{turn} \quad (8)$$

$$\begin{cases} \psi(l_{i,n-1}, l_{i,n}) = |\theta_{i,n+1} - \theta_{i,n}| \times \frac{180}{\pi} \\ \text{s.t. } \begin{cases} \theta_{i,n+1} = \text{atan} \left| \frac{y_{i,n+1} - y_{i,n}}{x_{i,n+1} - x_{i,n}} \right| \\ \theta_{i,n} = \text{atan} \left| \frac{y_{i,n} - y_{i,n-1}}{x_{i,n} - x_{i,n-1}} \right| \end{cases} \end{cases} \quad (9)$$

where, $\psi(l_{i,n-1}, l_{i,n})$ is the angle variable from the line $l_{i,n-1}$ that connects grid $p_{i,n-1}$ and $p_{i,n}$ to the line $l_{i,n}$ that connects grid $p_{i,n}$ and $p_{i,n+1}$, as shown in Fig. 1 (c); N_{turn} is the sum of turns between the target T_i to T_j ; ε_1 and ε_2 are the weight coefficients of the sum of turning angle variables and turning times, respectively, and $\varepsilon_1 + \varepsilon_2 = 1$.

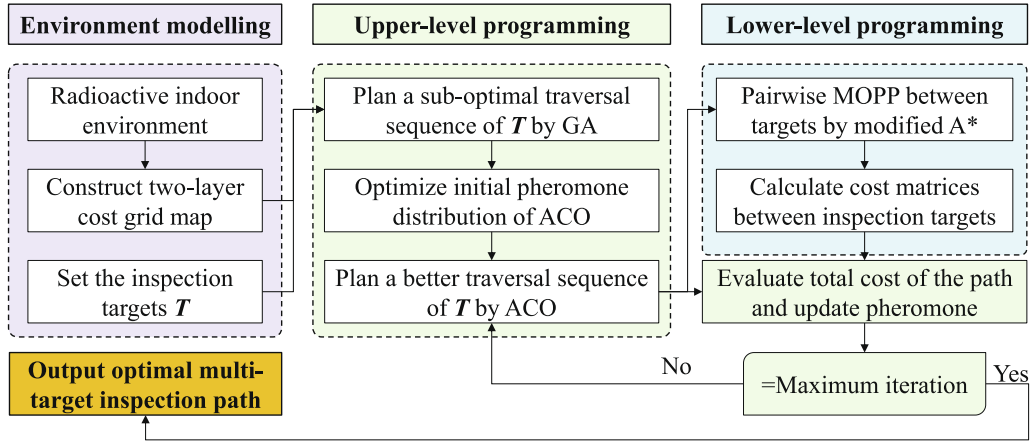


Fig. 3. The overall framework of environment modeling and ACO-GA-A*.

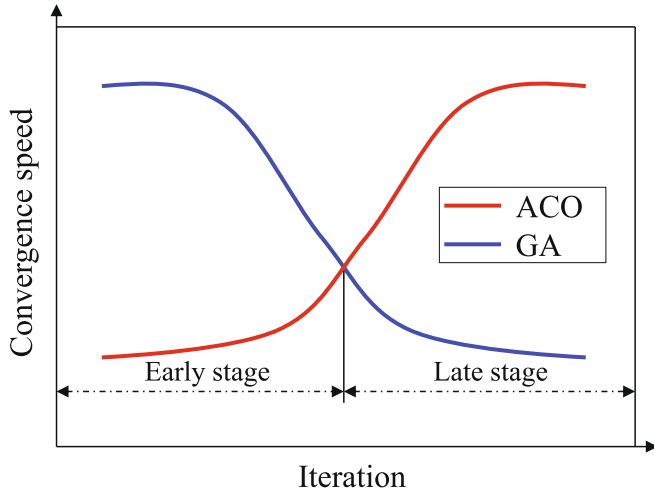


Fig. 4. The comparison of convergence speed between ACO and GA.

5. Methodology

To solve the MTIPP problem in complex radioactive indoor environment, a bi-level hybrid algorithm named ACO-GA-A*, combining ACO, GA and A*, is proposed. The overall framework of environment modeling and ACO-GA-A* is illustrated in Fig. 3. Firstly, a hybrid algorithm ACO-GA is developed in the upper-level programming to optimize the traversal sequence of inspection targets based on the cost matrices (cost graphs) of targets. Secondly, the modified A* in the lower-level programming generates the collision-free paths, considering multiple constraints between targets, thereby obtaining the cost matrices between the targets. The cost matrices obtained by the lower-level is then fed back to the upper-level to refine the traversal sequence, subsequently guiding the pair MOPP of lower-level. The detailed development of the bi-level hybrid algorithm is present in the following subsections.

5.1. ACO-GA for optimizing traversal sequence of inspection targets

The popular swarm intelligence algorithm ACO, developed by

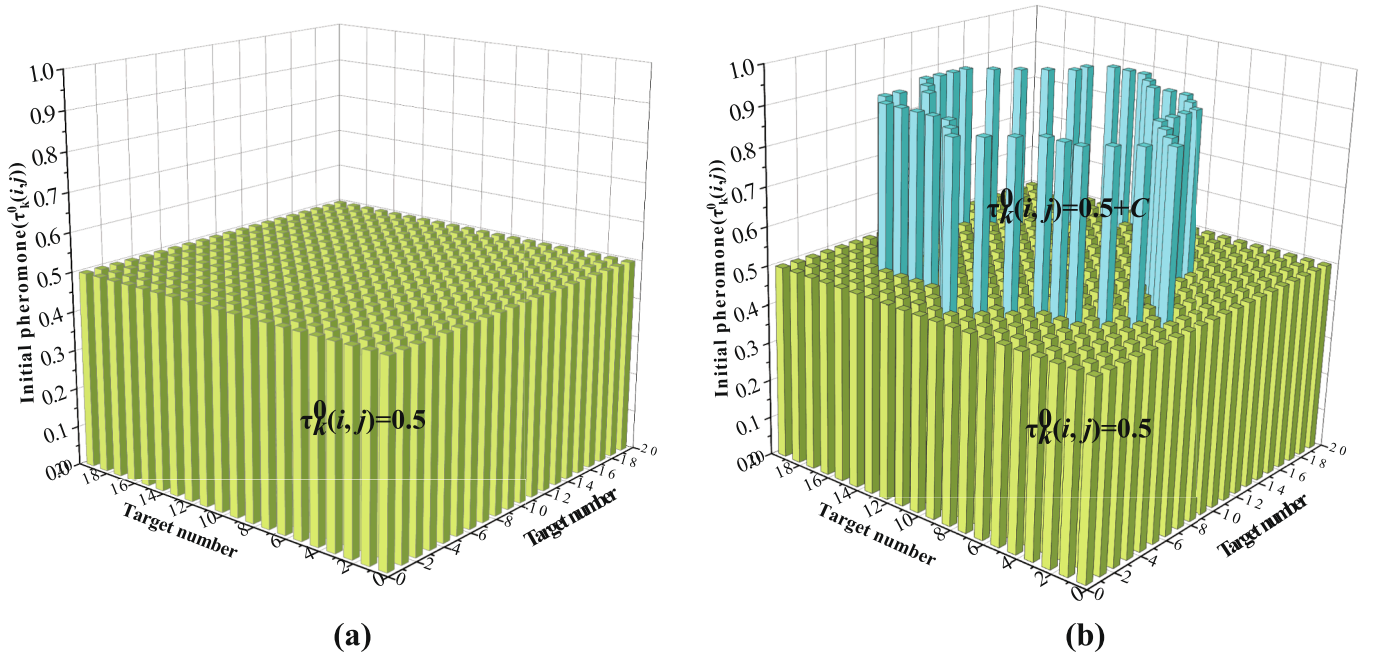


Fig. 5. The initial pheromone distribution of ACO. (a) uniform distribution of initial pheromone; (b) non-uniform distribution of initial pheromone based on GA.

Dorigo in 1970 (Dorigo et al., 1996), incorporates a positive feedback mechanism based on the state transition probability and pheromone update rules, inspired by the foraging behavior observed in real ant colonies. Contrary to the evolutionary algorithm GA, ACO initially exhibits a slower convergence speed due to the lack of pheromone guidance (Wen et al., 2021). Nevertheless, ACO's convergence speed gradually increases with iterations due to the positive feedback mechanism generated by pheromone deposition, a feature absent in GA. Therefore, ACO demonstrates superior search ability compared to GA in the late stage. Fig. 4 illustrates the comparison of convergence speed between ACO and GA. This characteristic has positioned ACO as more effective than GA and the simulated annealing (SA) algorithm in solving complex TSPs, particularly those with a large number of traveling targets (Cai, & Dong, 2007). Given this analysis, in this subsection, ACO serves as the primary framework of the hybrid algorithm, with a non-uniform initial pheromone distribution constructed by GA. Additionally, adaptive heuristic function and pheromone update strategies with elite selection are designed to enhance traditional ACO's ability to provide the traveling order of inspection targets by considering multiple constraints.

5.1.1. Non-uniform initial pheromone distribution based on GA

The initial pheromone of the traditional ACO is generally set to a uniform distribution, as shown as Fig. 5 (a), which will lead to blind search of the algorithm in the early stage, and thus reduce the convergence speed of the algorithm (Liu et al., 2023). To address this problem, a new initializing pheromone method based on the sub-optimal path obtained by GA is introduced to enhance the ability of the algorithm to guide ants to preferentially select the dominant optimal area. The method consists of two stages. In the first stage the randomness and rapidity of genetic is used to obtain a series of sub-optimal solutions (coarse search) to adjust the distribution of pheromones in ACO; the second stage is to use the ACO with positive feedback mechanism to solve the inspection path problem (accurate search).

GA is an evolutionary algorithm simulating natural selection, including the steps of encoding, fitness evaluation, selection, crossover, mutation and decoding. For solving TSP mentioned in this paper, the chromosome, gene and fitness in GA correspond to the candidate solutions (i.e., traveling order), optimization variables (i.e., inspection targets) and optimization objectives. In GA on TSP, each gene represents an inspection target. Moreover, the total number of genes in the chromosome is the number of all targets m . Hence, this problem can be transformed into a permutation and combination problem based on each cost matrix, which is obtained by the modified A* algorithm in Section 5.2. The detailed steps are as follows:

Step 1. Coding based on real number: deploying targets from 1 to m to indicate that there are m targets should be inspected by the MR.

Step 2. Tabu table creation: for each real gene code, the selected taboos list is labeled as 1, and the unselected taboos list is labeled as 0.

Step 3. Population initialization: set the evolutionary algebra counter G and the maximum evolutionary generation G_{max} , and generate initial population randomly.

Step 4. Individual evaluation: all permutation combinations are evaluated according to the fitness function $1/\mathcal{F}_q$, which can be calculated by Eq. (4).

Step 5. Individual selection: the selection operator is applied to the existing population, with individuals with good genes being retained for the next generation.

Step 6. Crossover operation: select individuals from the existing population based on crossover probability p_c . Select genes for crossover operation by crossover operators to generate new individuals.

Step 7. Mutation operation: individual are selected from the existing population based on the mutation probability p_m , and the mutation operators are used to mutate individual genes into alleles with alternative coding values.

Step 8. Evolution termination judgment: while $G \leq G_{max}$, repeats

Step 3-7. Otherwise, the evolution is terminated, and the individual with maximum fitness value in the population is the optimal solution, i.e., the sub-optimal traversal sequence of inspection targets $Path_{GA}$.

After applying GA to solve the TSP, a non-uniform distribution of initial pheromone setting is introduced to enhance guidance function for which dominant optional targets is prior selected in initial searching process. Specifically, the initial pheromone $\tau_k^0(i, j)$ on the edge between target T_i and target T_j is defined according the Eq. (10). And the traditional initial pheromone distribution and improved initial pheromone distribution are shown in Fig. 5.

$$\begin{cases} \tau_k^0(i, j) = 0.5 + C, & \text{if } i, j \in path_{GA} \\ \tau_k^0(i, j) = 0.5, & \text{else} \\ 0.5 \leq c \leq 1 \end{cases} \quad (10)$$

5.1.2. Adaptive heuristic function

To obtain the travel order of inspection targets, the solution procedure starts with a user-defined target, then select target by target and finally returns to the starting target. At t th iteration of algorithm, the next target T_j from the current target T_i of the path is selected by ant k among the untraveled targets recorded in J_k according the following pseudo-random scale rule:

$$s = \begin{cases} \text{argmax}_{j \in J_k} [\tau_k^t(i, j)]^\alpha [\eta_k^t(i, j)]^\beta, & \text{if } q \leq q_0 \\ P_k^t(i, j), & \text{else} \end{cases} \quad (11)$$

where,

$$P_k^t(i, j) = \begin{cases} \frac{[\tau_k^t(i, j)]^\alpha [\eta_k^t(i, j)]^\beta}{\sum_{u \in J_k} [\tau_k^t(i, u)]^\alpha [\eta_k^t(i, u)]^\beta}, & \forall j \in J_k \\ 0, & \text{else} \end{cases} \quad (12)$$

where, $\tau_k^t(i, j)$ is the pheromone between target T_i and T_j in the t th iteration; $\eta_k(i, j)$ is the heuristic function, which represents the total cost from T_i to T_j ; α and β are user-defined constant parameters that control the relative importance between the pheromone concentration factor and expected heuristic factor; q_0 is a constant parameter, and q denotes a random parameter in $[0, 1]$; $P_k^t(i, j)$ represents the transition probability from T_i to T_j for the k th ant in the t th iteration.

As a critical parameter affecting the state transition probability, the heuristic function determines the probability of the ant choosing the next inspection target, as shown in Eq. (12). Hence, the influence of path length, risk and energy consumption is considered into heuristic function. Meanwhile, an adaptive adjustment coefficient ξ is introduced to strengthen the impact of the heuristic function in the early stages while attenuating it in the later stages. This adaptive mechanism aims to balance the convergence speed and global search ability of the algorithm. The modified heuristic function can be seen in Eq. (13).

$$\begin{cases} \eta_k^t(i, j) = \xi \frac{1}{\omega_1 e_{ij}^L + \omega_2 e_{ij}^R + \omega_3 e_{ij}^E} \\ s.t. \xi = \begin{cases} \frac{I_{max} - I_c}{I_{max}}, & I_c \neq I_{max} \\ \frac{1}{I_{max}}, & I_c = I_{max} \end{cases} \end{cases} \quad (13)$$

where $(\omega_1 e_{ij}^L + \omega_2 e_{ij}^R + \omega_3 e_{ij}^E)$ represents the comprehensive evaluation value of the path between T_i and T_j obtained from Eq. (4); I_{max} and I_c represent the maximum and current iterations, respectively.

5.1.3. Pheromone update based on the elite strategy

To avoid falling into the local optimal solution prematurely due to excessive residual pheromones flooding the heuristic information, the

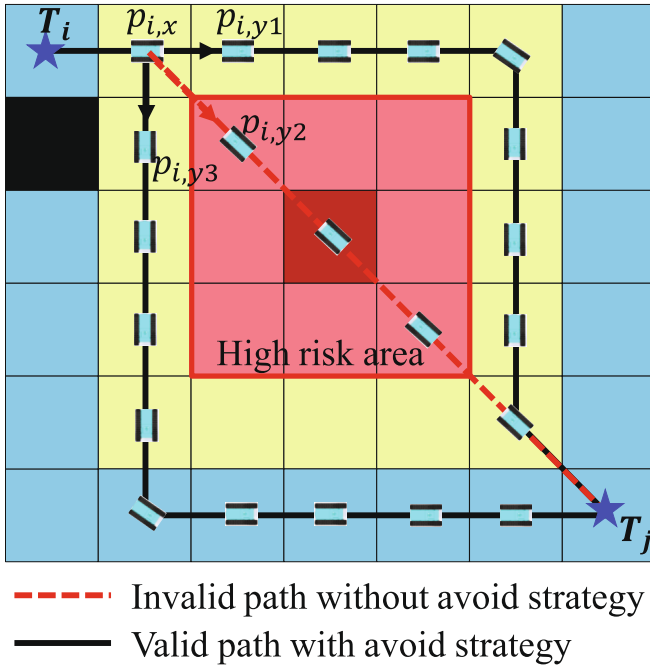


Fig. 6. The high-risk area avoidance strategy.

ACO algorithm needs to update pheromone after all the ants complete a cycle to improve the diversity of the ant colony population. In the field of PP applications, the pheromone updating rule of ACO is mainly affected by the path length according to Eq. (14).

$$\begin{cases} \tau^{t+1}(i,j) = (1-\rho)\tau^t(i,j) + \Delta\tau(i,j) \\ \Delta\tau(i,j) = \sum_k^{max} \Delta\tau_k(i,j) \\ s.t. \Delta\tau_k(i,j) \begin{cases} Q/L_k \\ 0, else \end{cases} \end{cases} \quad (14)$$

where, $\tau^{t+1}(i,j)$ and $\tau^t(i,j)$ are the pheromones on the path at iterations t and $t+1$, respectively; ρ is the pheromone volatilization factor within $[0,1]$; and Q is a constant that indicates the pheromone increment coefficient; $\Delta\tau_k(i,j)$ is the pheromone concentration released by the ant k on the path (i,j) between T_i and T_j ; max is the user-defined the maximum number of ants; L_k represents the path length of ant k from T_i to T_j .

ACO with single positive feedback mechanism has a good convergence speed. However, the excessive positive feedback pheromone concentrated in a good path will limits the exploration ability of the ant colony for the unknown paths, which leads the algorithm fall into a local optimum solution (Li, Xia, et al., 2020; Miao et al., 2021). Meanwhile, traditional pheromone updating rule only considering the influence of the path length, which cannot meet the requirement of the MTIPP under multi-factor constraints in this paper. Therefore, combining the positive and negative feedback mechanisms with multiple performance metrics, the pheromone updating rule are modified as following equation:

$$\begin{cases} \tau^{t+1}(i,j) = (1-\rho)\tau^t(i,j) + \rho \frac{Q}{T_{kmin}}, ij \in I_i \\ \tau^{t+1}(i,j) = (1-\rho)\tau^t(i,j) - \rho \frac{Q}{T_{kmax}}, ij \notin I_i \end{cases} \quad (15)$$

where, T_{kmin} and T_{kmax} are the multiobjective values of the optimal and worst inspection paths in t th iteration, which be calculated by Eq. (4); I_i is the set of inspection targets that minimize the objective function value of Eq. (2).

5.2. Modified A* for pairwise MOPP

The problem of searching pairwise PP can be described as ‘given a MR with two inspection target locations and a radioactive environment, find the least-cost collision-free path between targets’. Considering the significant time expenditure associated with iterative intelligent algorithms when dealing with PP problems, this paper adopts the A* algorithm to search for pairwise PP and introduces the following modifications: 1) The evaluation function is modified to corresponding to the objective function of the MOPP problem; 2) An avoidance strategy of high-risk area is designed to ensure the safety of MR.

5.2.1. Modified evaluation function

As a deterministic best-first search algorithm, the A* algorithm combines the characteristics of Dijkstra algorithm and Breadth-First Search (BFS) algorithm. It evaluates the current node by combining the cost $g(n)$ from the initial node to the current node n with the heuristic evaluation cost $h(n)$ from node n to the target node. The evaluation function can be described as following:

$$\begin{cases} f(n) = g(n) + h(n) \\ s.t. \begin{cases} g(n) = \sqrt{(x_n - x_s)^2 + (y_n - y_s)^2} \\ h(n) = \sqrt{(x_n - x_e)^2 + (y_n - y_e)^2} \end{cases} \end{cases} \quad (16)$$

where, (x_s, y_s) , (x_n, y_n) and (x_e, y_e) are the coordinates of the initial mode, current node and target node, respectively.

Considering multiple constraints of the pairwise PP problem, the actual cost function $g(n)$ and heuristic evaluation cost $h(n)$ should be modified as the weighted-sum cost in Eq. (2). As mentioned above, the waypoints between $T_i(p_{i,1})$ and $T_j(p_{i,|p_i|})$ are defined as $p_i = \{p_{i,1}, p_{i,2}, \dots, p_{i,|p_i|}\}$. Then, during the modified A* search, give a constructed path up to waypoint $p_{i,x}$, the path cost to each candidate waypoint $p_{i,y}$ adjacent to x can be calculated by the following formulation:

$$g(p_{i,y}) = g(p_{i,x}) + \omega_1 L(p_{i,x}, p_{i,y}) + \omega_2 R(p_{i,x}, p_{i,y}) + \omega_3 E(p_{i,x}, p_{i,y}) \quad (17)$$

where, $g(p_{i,x})$ is the total cost of the path up to the waypoint $p_{i,x}$; $L(p_{i,x}, p_{i,y})$, $R(p_{i,x}, p_{i,y})$ and $E(p_{i,x}, p_{i,y})$ from $p_{i,x}$ to $p_{i,y}$ can be calculated by Eq. (6)-(8). The heuristic function $h(p_{i,x})$ for evaluating the cost from waypoint $p_{i,x}$ to the target $p_{i,|p_i|}$ can be defined as Eq. (18).

$$h(p_{i,x}) = \omega_1 L(p_{i,x}, p_{i,|p_i|}) + \omega_2 R(p_{i,x}, p_{i,|p_i|}) + \omega_3 E(p_{i,x}, p_{i,|p_i|}) \quad (18)$$

It is worth explaining that the Eq. (17) and (18) is different from the Eq. (3) and (4). The former two formulas are developed to guide the A* algorithm in searching for a better point from candidate waypoints to form an optimal path, while the last two equations are employed to evaluate the quality of a path.

5.2.2. Avoidance strategy of high-risk areas with intolerable radiation dose rates

As mentioned in Section 3.2, risk avoidance is an essential optimize objective of the inspection path, aiming to prevent MRs from traversing areas with radiation dose rates exceeding the tolerance of sensitive electronic components equipped on the MRs (Sharp, & Decréton, 1996). Therefore, a high-risk avoidance strategy is incorporated into the path search of the modified A* algorithm to circumvent the potential failure of the MR while navigating through areas with intolerable radiation dose rates. The avoidance strategy is designed as shown as Fig. 6. The detailed process unfolds as follows: Upon arrive at the current waypoint, the on-board radiation detector of the MR detects the radiation dose rates of passable adjacent waypoints. Subsequently, the radiation dose

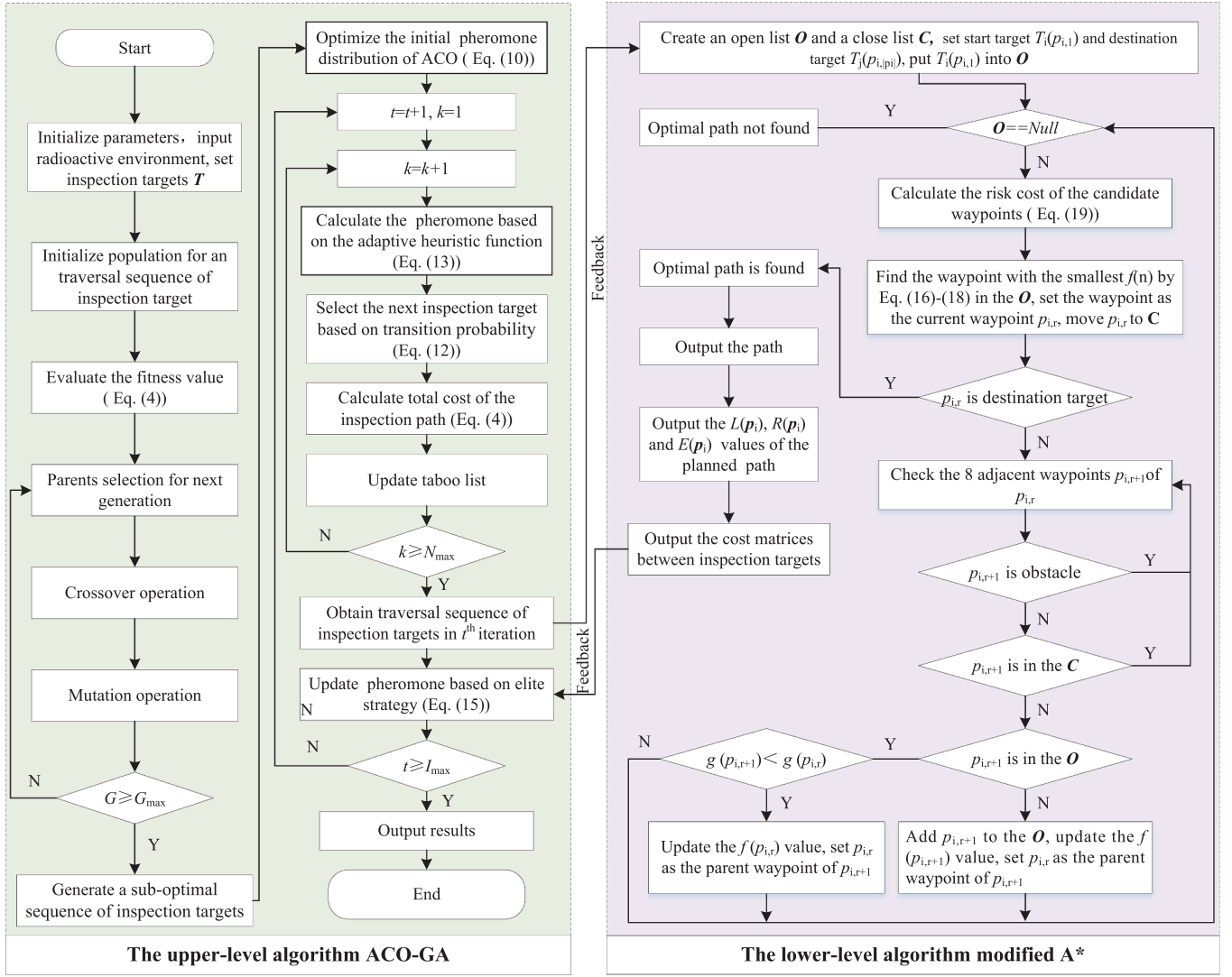


Fig. 7. The flowchart of the bi-level hybrid algorithm ACO-GA-A*.

rates, representing risk degree cost, of adjacent waypoints are redefined using Eq. (19). As shown in Fig. 6, the current waypoint $p_{i,x}$ has three passable adjacent waypoints $p_{i,y1}$, $p_{i,y2}$ and $p_{i,y3}$. Among them, $p_{i,y2}$ is situated in a high-risk area. Therefore, the radiation dose rate of $p_{i,y2}$ is set to infinity, while the radiation dose rates of the other adjacent waypoints are assigned as the actual calculated values.

$$R(p_{i,y}) = \begin{cases} R(p_{i,y}), & \text{if } R(p_{i,y}) < RI_{max} \\ Inf, & \text{else} \end{cases} \quad (19)$$

The general framework of the modified A* algorithm for solving pairwise MOPP from target $T_i(p_{i,1})$ to target $T_j(p_{j,|p_i|})$ can be described as follows:

Step 1. Initialization: set open *open list* (*O*) for recording the waypoints that have been generated but not visited and a *close list* (*C*) for storing the waypoints that have been visited; put starting waypoint $p_{i,1}$ into *O* and destination waypoint $p_{j,|p_i|}$ into *C*; $f(p_{i,1})=0$; Set RI_{max} .

Step 2. Waypoint selection: calculate the risk cost of the candidate waypoints by Eq. (19); candidate waypoint $p_{i,r}$ with minimal f -value in *O* is selected as the next waypoint; delete $p_{i,r}$ from *O*, and stored it into *C*.

Step 3. Stop criteria: algorithm stops if the target $T_j(p_{j,|p_i|})$ is reached in Step 2.

Step 4. Search extension: for each waypoint $p_{i,r+1}$ adjacent to $p_{i,r}$ by

edge $(p_{i,r}, p_{i,r+1})$ in the graph, (a) If $p_{i,r+1} \in C$, check another candidate waypoint; (b) If $p_{i,r+1} \notin (O \cup C)$, add the waypoint $p_{i,r+1}$ into the *O*; (c) If $(p_{i,r+1} \notin C) \cap (p_{i,r+1} \in O)$, calculate f -value $f'(p_{i,r+1})$ of the path from $p_{i,r}$ to $p_{i,r+1}$ and compare it with the previous value of $f(p_{i,r+1})$ of $p_{i,r+1}$. If $f'(p_{i,r+1}) < f(p_{i,r+1})$, set $f(p_{i,r+1}) = f'(p_{i,r+1})$ and update the precursor waypoint of the path waypoint $p_{i,r+1}$ to $p_{i,r}$.

Based on the aforementioned algorithm improvements, the bi-level hybrid algorithm ACO-GA-A* is developed. The flowchart of the ACO-GA-A* is depicted in Fig. 7. The pseudo code of MTIPP process based on the ACO-GA-A* algorithm is shown in Table 2.

6. Experiments and result analysis

To demonstrate the performance of the proposed bi-level programming framework and its composed algorithm, extensive simulation experiments and real environment experiment are performed. The simulated experiments include the effectiveness verification experiment of the upper-level algorithm ACO-GA, the superiority of the modified A* in the low-level programming and the performance of ACO-GA-A* in solving MTIPP problems. Finally, an experiment in real environment is performed to verify the MTIPP approach. All comparison algorithms are coded in C#, the simulation experiments are executed on a 64-bit Windows 10 desktop with Intel(R) Xeon(R) Platinum 8383C 2.70 GHz

Table 2

The pseudo code of ACO-GA-A* in solving MTIPP problems.

Algorithm 1 Bi-level hybrid algorithm ACO-GA-A* algorithm for MTIPP

Input: Radioactive indoor environment model based on two-layer cost grid map; m inspection targets $T = \{T_1, T_2, \dots, T_m\}$.

1. Initialize the parameters: parameters of ACO I_{max} , N_{max} , $\tau_k^0(i, j)$, α , β , γ ; parameters of GA: G_{max} , n_p , p_c , p_m , gap ; cost weight coefficients: ω_1 , ω_2 and ω_3 . Set T_1 as the start inspection target.
2. Construct an *open list* (O) and a *close list* (C) and set start target $T_i(p_{i,1})$ and destination target $T_{i+1}(p_{i,p_i})$
3. Set $O \leftarrow p_{i,1}$, $C \leftarrow p_{i,p_i}$
4. Add candidate waypoints $p_{i,y}$ around $p_{i,1}$ into O , remove $p_{i,1}$ from the O and add it into the C
5. **while** both $O \neq \emptyset$ and $C \neq \emptyset$ **do**
6. Determine whether the candidate waypoints are located in the risk area with high radiation dose rate and calculate the risk cost by Eq. (19).
7. Find the candidate waypoint $p_{i,2}$ with the smallest $f(n)$ value in the O based on the Eq. (16)-(18).
8. Remove $p_{i,2}$ from O and add $p_{i,2}$ into the C
9. According to the 8 adjacent waypoints of $p_{i,2}$, update the O and the C
10. **end while**
11. **return** (optimal pairwise path, cost matrices)

/********Modified A* in lower level** *****/

12. **Function GA**

13. Initialize the solutions and associate them with individuals in current population
14. Evaluate population using fitness function $1/\mathcal{F}_q$ calculated by Eq. (4).
15. **while** ($G \leq G_{max}$)
16. Select parents
17. Crossover parents to create offspring, retain the individuals with greater fitness with a ratio of p_c
18. Mutation operation, the remained individuals are produced the next generation based on a ratio of p_m
19. **end while**
20. **return** the sub-optimal traversal sequence of inspection targets

21. **Function IACO**

22. Initialize pheromone matrix $\tau_k^0(i, j)$ based on **Function GA** and Eq. (10)
23. **while** $t = 1$ to I_{max} **do**
24. **for** $k \leftarrow 1, 2, \dots, N_{max}$ **do**
25. Put all ants in T_1
26. **while** ant k does not traverse all detection targets and returns T_1
27. Calculate the probabilities of all untraveled inspection targets according to the Eq. (12) and Eq. (13), and select the next inspection target by the probabilistic mechanism
28. **else**
29. Calculate total cost of the planned inspection path by Eq. (4)
30. $k \leftarrow k + 1$
31. **while** all ants traverse all detection targets and return T_1 **do**
32. Select the optimal inspection path by Eq. (3)
33. Calculate the cost matrices by the **Modified A*** and update the pheromone by Eq. (15)
34. **end while**
35. $t \leftarrow t + 1$
36. **Output** (optimal multi-target inspection path, traversal sequence of the inspection targets and total cost of the inspection path)

Table 3

Basic information of TSP examples for resolving.

TSP	Number of cities(n)	Number of edges($n(n-1)/2$)	BKS
Att48	48	1128	33,522
Ei151	51	1275	426
Eil76	76	2850	538
KroA100	100	4950	21,282
Pr136	136	9180	96,772
Ch150	150	11,175	6528
U159	159	12,561	42,080
KroA200	200	19,900	29,368

Table 4

Parameters of the comparison algorithms in solving TSP examples.

Algorithms	Parameters
ACO-GA	$\alpha = 1, \beta = 5, \rho = 0.1, Q = 1000, p_c = 0.9, p_m = 0.05, Gap = 0.9, I_{max} =$
GA-ACO	$500, N_{max} = 200, G_{max} = 500, n_p = 200$
ACO	$\alpha = 1, \beta = 5, \rho = 0.1, Q = 1000, I_{max} = 500, N_{max} = 200$
GA	$p_c = 0.9, p_m = 0.05, Gap = 0.9, G_{max} = 500, n_p = 200$
PSO	$N_{max} = 500, n = 1000, \omega = 0.8, c_1 = 0.5, c_2 = 0.5$

CPU, 128G memory, and simulation software is MATLAB R2020b (Math Works).

6.1. Effectiveness of the upper-level algorithm ACO-GA

To verify the effectiveness of the hybrid algorithm ACO-GA in optimizing traversal sequence of inspection targets, TSP examples of different sizes in TSPLIB (Reinelt, 1991) are used for verification. TSP examples of scales from 48 to 200 are selected from TSPLIB for resolving. The basic information of these TSP examples is shown in Table 3. BKS denotes the length of the best-known solution of each TSP examples.

Under the same preconditions, we did the experiments for the same TSP examples with the proposed ACO-GA, hybrid algorithm GA-ACO (Liu et al., 2022), meta-heuristic algorithms ACO, GA and PSO and the experimental results are compared. The parameters of comparison algorithms are set in Table 4. And the results are listed in Table 5. *Best* and *worst* represent the maximum and minimum length among ten solutions for each TSP example, respectively. *Average* and *SD* denote the average value and standard deviation computed based on ten solutions, respectively. *Error*(%) denotes the relative error of *Average* with respect to BKS, which calculated by Eq. (20). And the *PE*(%) represents the relative error

Table 5

Experimental results of the comparison algorithms in solving TSP examples.

Algorithm	TSP	Att48	Ei151	Eil76	KroA100	Pr136	Ch150	U159	KroA200
	<i>BKS</i>	33,522	426	538	21,282	96,772	6528	42,080	29,368
ACO-GA	<i>Best</i>	33,524	431	545	21,298	98,516	6628	42,541	30,051
	<i>Worst</i>	33,845	438	558	22,447	101,569	6823	44,476	31,111
	<i>Average</i>	33,611	434	551	21,882	100,094	6696	43,646	30,519
	<i>SD</i>	86.12	2.27	4.27	312.93	1033.66	58.12	527.84	305.98
	<i>Error</i> (%)	0.27	1.88	2.47	2.82	3.43	5.17	3.72	3.92
GA-ACO	<i>PE</i> (%)	0.26	0.81	1.18	2.74	1.60	1.01	2.60	1.56
	<i>Best</i>	33,601	431	549	21,883	102,413	6788	44,343	31,378
	<i>Worst</i>	35,444	451	586	23,519	107,244	7109	46,751	33,333
	<i>Average</i>	34,219	439	567	22,443	104,469	6923	45,516	32,265
	<i>SD</i>	514.64	5.22	10.39	423.10	1565.73	91.05	582.02	522.78
ACO	<i>Error</i> (%)	2.08	3.09	5.43	5.46	7.95	6.04	8.16	9.86
	<i>PE</i> (%)	1.84	1.81	3.32	2.56	2.01	1.98	2.64	2.83
	<i>Best</i>	34,425	443	565	22,388	106,873	6753	44,899	31,323
	<i>Worst</i>	35,251	449	571	22,709	108,837	6827	45,474	31,873
	<i>Average</i>	34,750	446	569	22,491	107,998	6800	45,235	31,615
GA	<i>SD</i>	231.10	2.40	1.32	96.48	681.16	19.40	196.41	157.27
	<i>Error</i> (%)	3.66	4.60	5.67	5.68	11.60	4.16	7.50	7.65
	<i>PE</i> (%)	0.94	0.64	0.54	0.46	1.05	0.69	0.75	0.93
	<i>Best</i>	33,608	439	585	23,020	108,066	12,371	94,889	104,001
	<i>Worst</i>	37,340	470	660	29,817	204,613	17,834	134,384	146,048
PSO	<i>Average</i>	35,008	451	619	26,710	174,042	16,312	115,747	118,824
	<i>SD</i>	1058.14	7.92	20.83	2077.33	22944.74	1396.93	13100.90	11105.12
	<i>Error</i> (%)	4.43	5.78	15.14	25.51	79.85	149.88	175.06	304.60
	<i>PE</i> (%)	4.17	2.59	5.85	16.03	61.05	31.86	21.98	14.25
	<i>Best</i>	33,834	437	567	22,361	102,641	7438	49,252	39,376
PSO	<i>Worst</i>	35,492	462	608	25,584	114,023	8073	57,822	49,304
	<i>Average</i>	34,767	447	585	24,319	107,539	7767	51,817	44,387
	<i>SD</i>	493.00	7.64	12.96	960.49	3888.82	175.57	2275.76	3234.02
	<i>Error</i> (%)	3.71	5.00	8.80	14.27	11.13	18.98	23.14	51.14
	<i>PE</i> (%)	2.76	2.37	3.18	8.76	4.77	4.42	5.21	12.73

of *Average* with respect to *Best*, which calculated by Eq. (21).

$$Error(\%) = \frac{Average - BKS}{BKS} \times 100\% \quad (20)$$

$$PE(\%) = \frac{Average - Best}{Best} \times 100\% \quad (21)$$

The results presented in Table 5 indicate that the solutions computed with ACO-GA for these TSP examples are obviously better than those obtained by other algorithms. Notably, while ACO-GA did not find *BKS* of corresponding TSP examples in ten tests, the *Error*(%) of the solutions is controlled within 4 %, which is smaller than other algorithms. For larger-scale TSP examples, such as Pr136, Ch150, U159, and KroA200, *Error*(%) computed by other algorithms exceed 5 %, and some even exceed 10 %.

Fig. 8 shows the evolutionary process of the *Best* computed with the five algorithms for these TSP examples. For each example in Fig. 8, the solution computed by ACO-GA is superior to that based on other algorithms at the same stage, and can converge to the *Best* within the smaller number of iterations. The best solutions of these TSP examples computed by ACO-GA are demonstrated in Fig. 9. It can be seen from the Fig. 9 that the traversal order formed by ACO-GA is good, and only the planned path of large-scale example KroA200 cross once. The above analysis results show that the hybrid algorithm ACO-GA can provide a better solution for optimizing the traversal order of targets.

6.2. Construction of simulated radioactive environment

To investigate the effectiveness of the ACO-GA-A*, seven hypothetical scenarios with differences in the information of dimension, radioactive sources and obstacles are constructed based on the two-layer cost grid map, as shown in Fig. 10.

The scenarios 1–3 depict two inspection targets surrounded by environments of varying complexity, as illustrated in Fig. 10(a)–(c). These scenarios are utilized to evaluate the superiority of the modified A* for

pairwise MOPP in the lower-level programming. Among them, scenario 1 illustrates a situation where the distance between the two targets is short, and the surrounding radioactive sources and obstacles are sparse. Scenario 2 presents a scenario where the distance between the two targets is short, and there are few radioactive sources but dense obstacles around them. Scenario 3 depicts a scenario where the distance between the two targets is long, and the surrounding radioactive sources and obstacles are dense. In addition, the scenarios 4–7 shown in Fig. 10 (d)–(g) are constructed to assess the effectiveness of the proposed hybrid algorithm in solving MTPP under various complexities of radioactive indoor environments. Scenario 4 displays a small-scale factory building with multiple radioactive sources of different intensity and sparse obstacles. Scenario 5 describes a medium-scale factory building with multiple radioactive sources of equal intensity and sparse obstacles. Scenario 6 demonstrates a medium-scale factory building with multiple radioactive sources of equal intensity and dense obstacles. Scenario 7 portrays a large-scale factory building with multiple radioactive sources of different intensity and dense obstacles. The detailed characteristics of the hypothetical scenarios are described in Table 6.

6.3. Superiority of the modified A* in the lower-level programming

As mentioned above, the goal of the lower-level programming is to provide a well-estimated cost graph of targets through the MOPP between each target. Many non-deterministic algorithms, such as ACO and GA (Lai, & Smith, 2022), have been applied to plan paths for MRs in complex radioactive environments. To verify the superiority of modified A* algorithm in solving the pairwise MOPP problem, original A*, ACO, GA and modified A* are employed to plan paths under scenarios 1–3. For fairness, the objective function of ACO and the fitness function of GA are modified as Eq. (2). To minimize the impact of randomness inherent in population-based intelligent algorithms on the comparison results, each algorithm is executed 10 times. The mean value and standard deviation of the results (i.e., path length *L*, risk degree *R*, turning energy consumption *E*, total cost *M* and computation time *CT*) under different

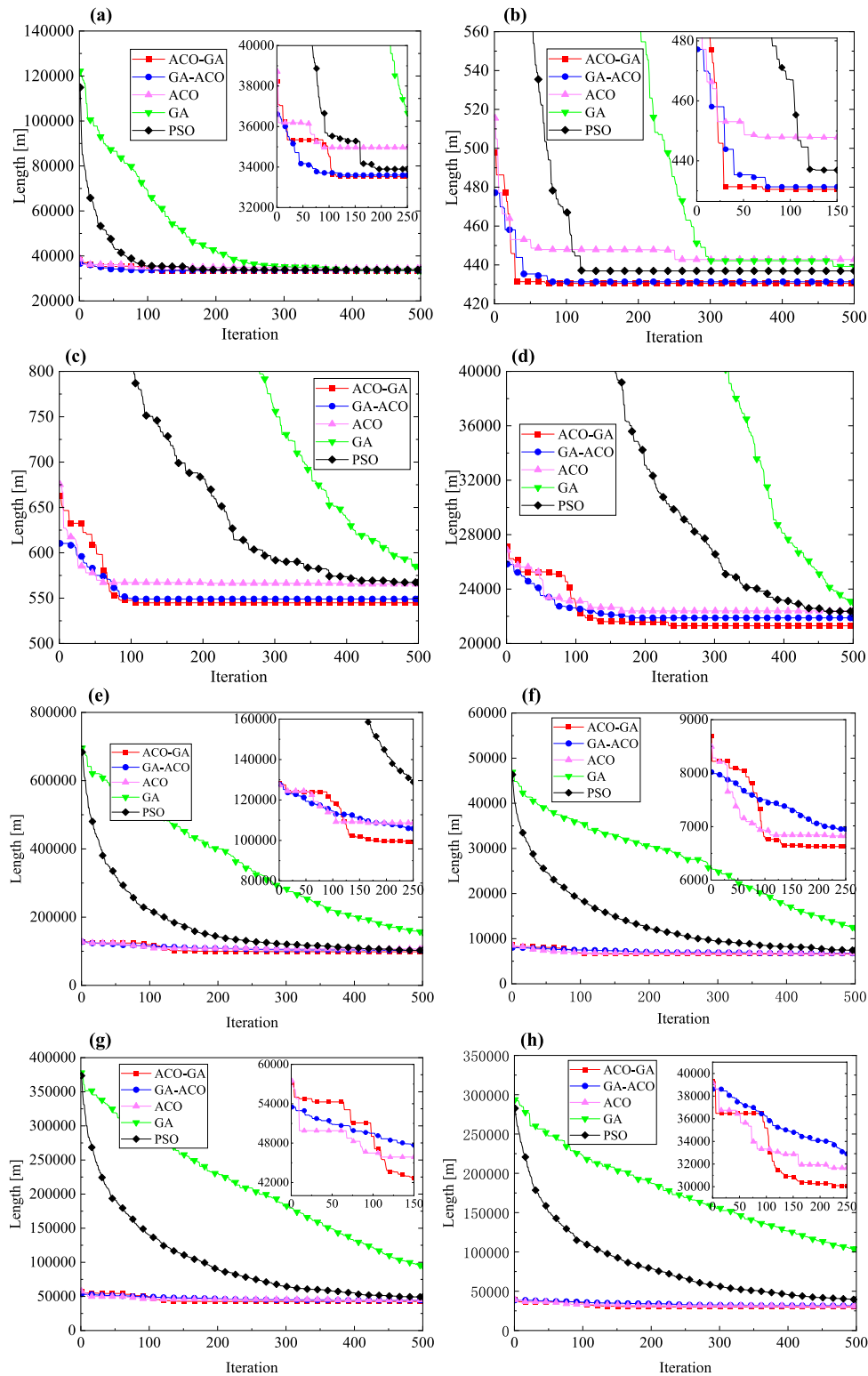


Fig. 8. The evolutionary process of the best solutions. (a) Att48; (b) Ei151; (c) Eil76; (d) KroA100; (e) Pr136; (f) Ch150; (g) U159; (h) KroA200.

scenarios are shown in Fig. 11. The optimal pairwise paths planned by the four algorithms are shown in Fig. 12.

The following conclusions can be drawn from the analysis of Fig. 11: 1) The modified A*, incorporating a multiple constraint evaluation function, outperforms the original A* algorithm significantly in terms of total cost. This makes it more suitable for solving MOPP problems; 2) In terms of path length, the performance of A* in scenarios 1 and 2 is close

to the modified ACO and GA, but it performs better in scenarios 3. For risk degree, the modified A* performs better than other algorithms in all three scenarios. In the context of turning energy consumption, the performance of the modified A* in scenario 1 and 3 is significantly better than the modified ACO and GA. Regarding as the total cost based on the weight coefficient, the modified A* is better than that of the modified ACO and GA. Especially in terms of time, the deterministic algorithm A*

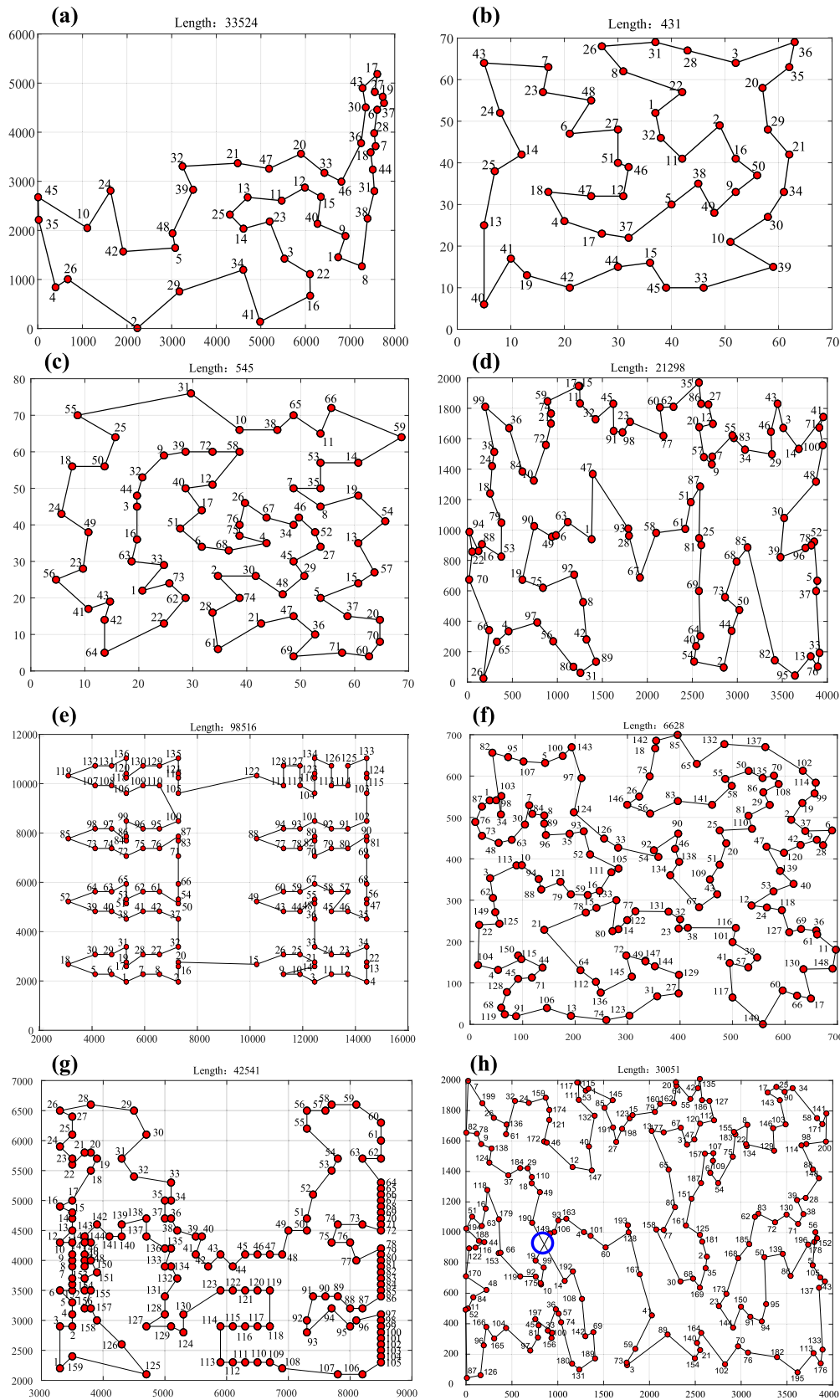


Fig. 9. The best solution computed by ACO-GA. (a) Att48; (b) Ei151; (c) Ei176; (d) KroA100; (e) Pr136; (f) Ch150; (g) U159; (h) KroA200.

has obvious advantages over ACO and GA; 3) With the increase of the complexity of the surrounding environments, the robustness of modified GA will be enhanced, while modified ACO will be weakened, which can

be seen from the standard deviations of different evaluation indexes. The robustness of the modified A* is not be affected by the change of the environment. It is can be seen from Fig. 12, the optimal path obtained by

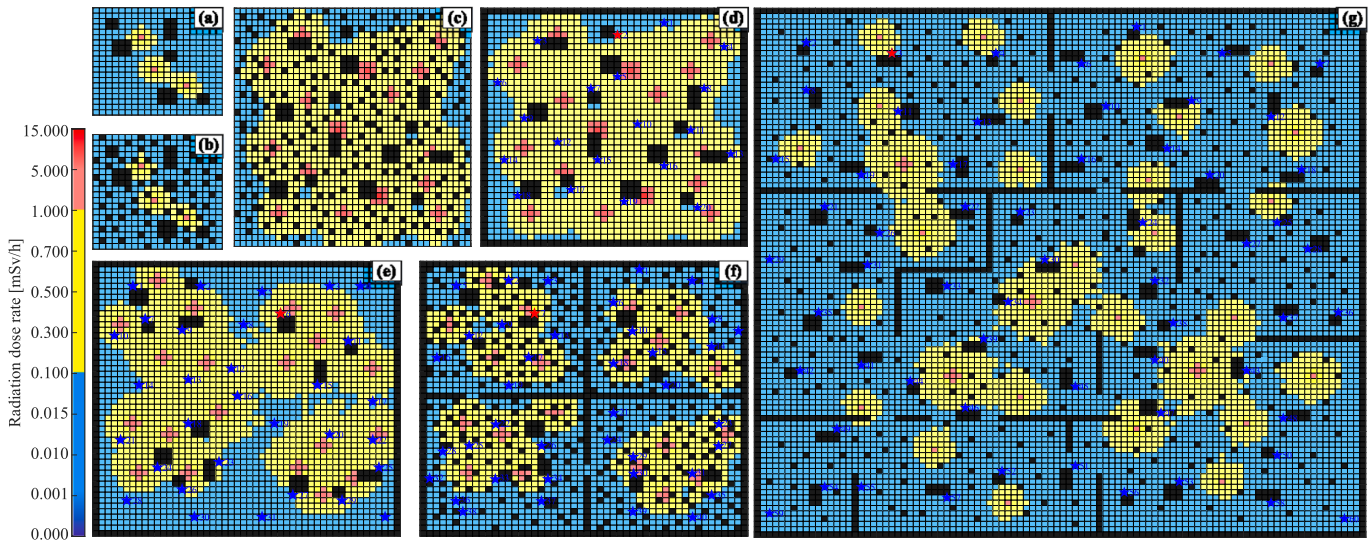


Fig. 10. Hypothetical radioactive indoor environments. (a) scenario 1; (b) scenario 2; (c) scenario 3; (d) scenario 4; (e) scenario 5; (f) scenario 6; (g) scenario 7.

Table 6

Characteristics of the hypothetical scenarios.

ID	Dimension (B × L × H) / mm ³	Radiation sources				Obstacles	Target number
		Number	Energy (MeV)	Activity (GBq)	Location (x (m), y (m))		
1	20 × 20 × 5	3	0.25 of all	1.12	(7.5, 14.5), (9.5, 8.5), (14.5, 5.5)	Sparse	—
2	20 × 20 × 5					Dense	—
3	40 × 40 × 5	16	1.25, 1.25,	1.12	(11.5, 35.5), (34.5, 34.5), (5.5, 30.5), (22.5, 29.5), (31.5, 29.5),	Dense	—
4	40 × 40 × 5		1.25, 2.5, 1.25, 1.25, 1.25, 2.5, 1.25, 1.25, 1.25, 1.25, 3.75, 1.25, 1.25, 1.25	1.12	(12.5, 24.5), (26.5, 25.5), (17.5, 19.5), (6.5, 16.5), (33.5, 15.5), (12.5, 12.5), (32.5, 10.5), (24.5, 8.5), (7.5, 5.5), (18.5, 5.5), (34.5, 5.5)	Sparse	20
5	50 × 50 × 5	20	1.25 of all	1.12	(7.5, 11.5), (15.5, 11.5), (6.5, 20.5), (12.5, 18.5), (20.5, 22.5), (33.5, 12.5), (33.5, 7.5), (37.5, 16.5), (42.5, 10.5), (46.5, 18.5), (4.5, 38.5), (10.5, 46.5), (11.5, 32.5), (16.5, 41.5), (18.5, 31.5), (31.5, 31.5), (32.5, 41.5), (36.5, 35.5), (39.5, 40.5), (45.5, 30.5)	Sparse	40
6	50 × 50 × 5	20			(20.5, 94.5), (37.5, 94.5), (63.5, 90.5), (84.5, 89.5), (20.5, 81.5), (43.5, 82.5), (69.5, 79.5), (88.5, 76.5), (7.5, 73.5), (24.5, 70.5), (43.5, 69.5), (64.5, 67.5), (84.5, 63.5), (27.5, 57.5), (65.5, 57.5), (52.5, 51.5), (17.5, 43.5), (45.5, 44.5), (59.5, 41.5), (78.5, 41.5), (32.5, 30.5), (72.5, 31.5), (91.5, 30.5), (17.5, 24.5), (43.5, 26.5), (62.5, 20.5), (80.5, 21.5), (29.5, 16.5), (41.5, 6.5) , (76.5, 10.5)	Dense	40
7	100 × 100 × 5	30	1.0 of all	1.12		Dense	60

the path planners (i.e., original A*, modified ACO and modified GA) without avoidance strategy are prone to pass through the high-risk zones (the red areas with dose rate greater than 1 mSv/h). Additionally, the modified A* can generate a path with a lower total cost compared with the compared algorithms. It can be seen that from the perspective of practicality, the modified A* algorithm has a significantly higher time efficiency than general intelligent algorithms while ensuring solution quality.

The MTIPP problem involved in this paper belongs to the symmetric TSP, so the costs of travelling from inspection target T_i to T_j is equal to the costs of travelling from inspection target T_j to T_i . Therefore, cost matrices of different environments calculated by the modified A* is symmetric matrices, which can be visualized cost graphs. Figs. 13–16 show the cost graphs of scenarios 4–7 obtained by modified A*. These cost graphs provide cost matrix data for the upper-level programming that discussed in the following section.

6.4. Performance of ACO-GA-A* for solving MTIPP problem

In this section, the performance of the proposed bi-level hybrid algorithm ACO-GA-A* in terms of sensitivity, reliability and convergence

is verified by comparing with other algorithms in four different complexity of radioactive scenarios (i.e., scenario 4–7). For the lack of ready-made bi-layer algorithm for MTIPP problem in radioactive environment, we have not found existing algorithm validate the effectiveness of our proposed algorithm. For fair comparison, we have combined some traditional intelligent algorithms and its improved variants for MTIPP with modified A* algorithm. The comparison algorithms are described as following: 1) GA-A*: the traditional GA (Lai, & Smith, 2022) combined with modified A*; 2) ACO-GA: the ACO (Tang et al., 2022; Xie et al., 2022) combined with modified A*; 3) GA-ACO-A*: the skeleton of the algorithm is GA, whose initial population is enhanced by ACO (Liu et al., 2022), and modified A* is incorporated; 4) ACO-GA-A*: our proposed algorithm. All the algorithms are use the same path evaluation function defined in Eq. (2). The parameters of these algorithms are set in Table 7.

6.4.1. Sensitivity analysis

Since the MTIPP of MRs in this paper considers three constraints of path length, risk degree and energy consumption, it is necessary to verify that the optimal path of user preference can be found by adjusting cost coefficient weights (ω_1 , ω_2 and ω_3). To this end, this subsection com-

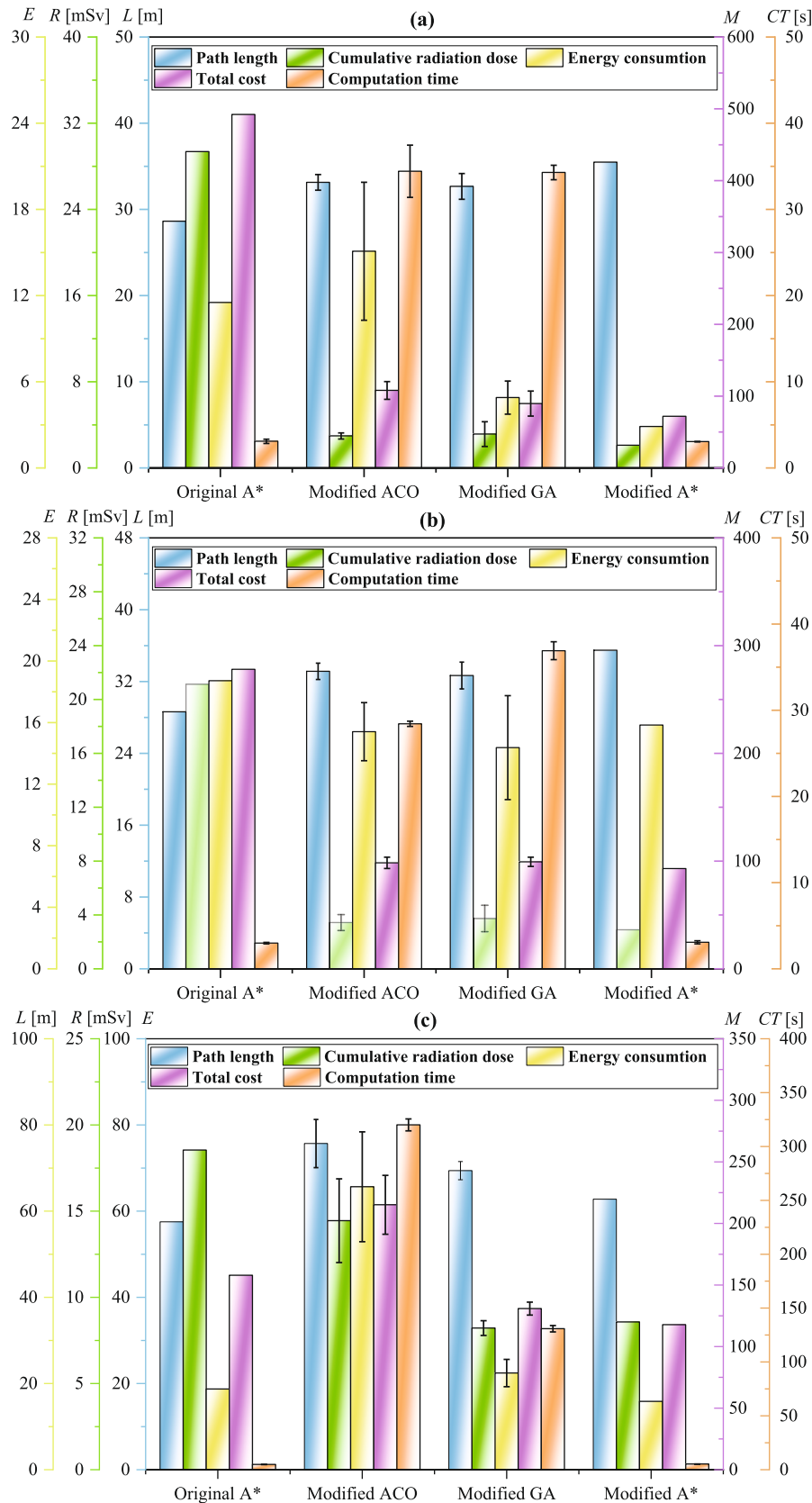


Fig. 11. Mean value and standard deviation of the results obtained by different algorithms under different scenarios. (a) scenario 1; (b) scenario 2; (c) scenario 3.

pares the sensitivity of the proposed algorithm with other algorithms for four groups of weights according to the requirements of actual working conditions for MRs in radioactive environment. In which, the (1, 0, 0)

represents the shortest path, the (0, 1, 0) shows the safest path with minimum cumulative radiation dose, the (0, 0, 1) displays the path with the minimum turning energy consumption and the (1, 1, 1) denotes the

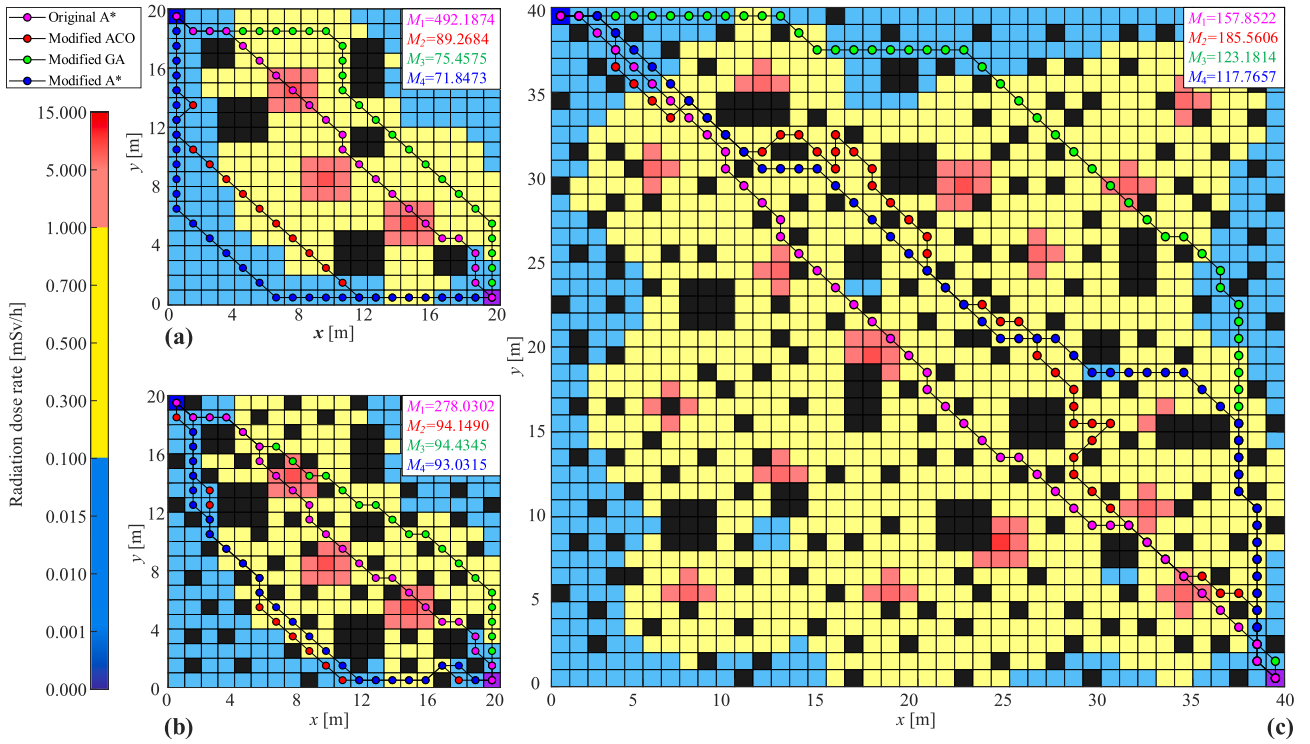


Fig. 12. The optimal pairwise paths planned by the four algorithms under different scenarios. (a) scenario 1; (b) scenario 2; (c) scenario 3. M_1 , M_2 , M_3 and M_4 represent the total cost of the optimal path planned by the original A*, modified ACO, modified GA and modified A*, respectively.

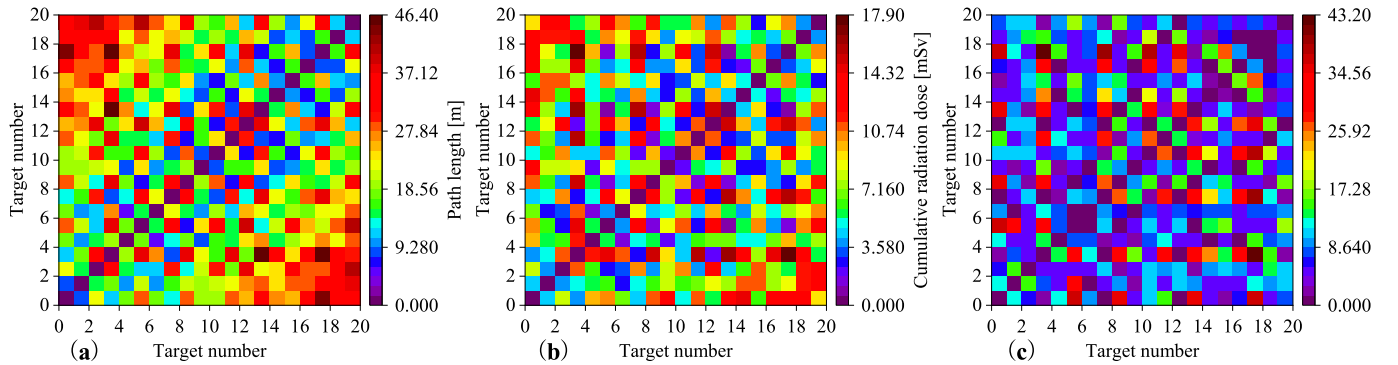


Fig. 13. Cost graphs of scenario 4 obtained by modified A*. (a) path length; (b) risk degree; (c) turning energy consumption.

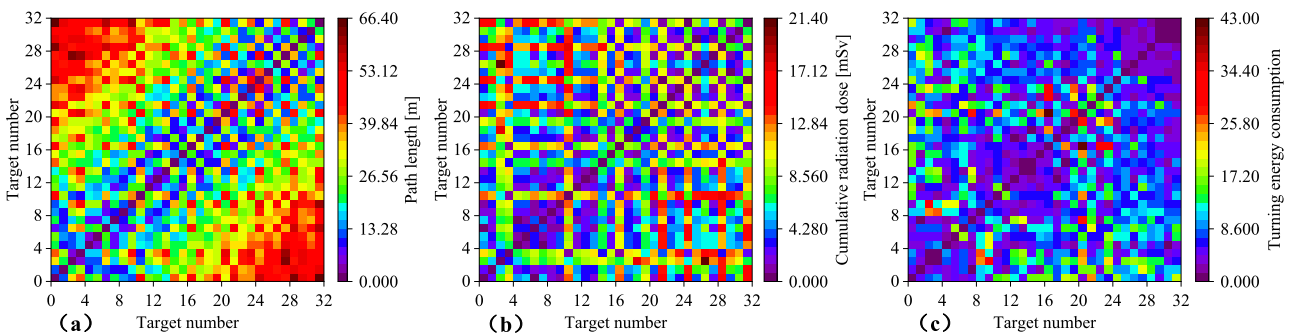


Fig. 14. Cost graphs of scenario 5 obtained by modified A*. (a) path length; (b) risk degree; (c) turning energy consumption.

optimal path considering all the three constraints simultaneously. Each algorithm with different weights runs 20 times under four scenarios of different complexity (i.e., scenarios 4–7). The boxplots of evaluation indexes obtained by different algorithms under different scenarios are

shown in Figs. 17–20. And, the shortest path, safest path and minimum energy consumption path planned by our proposed algorithm under different scenarios are shown in Figs. 21–24.

As depicted in Figs. 14–17: 1) For scenarios with different

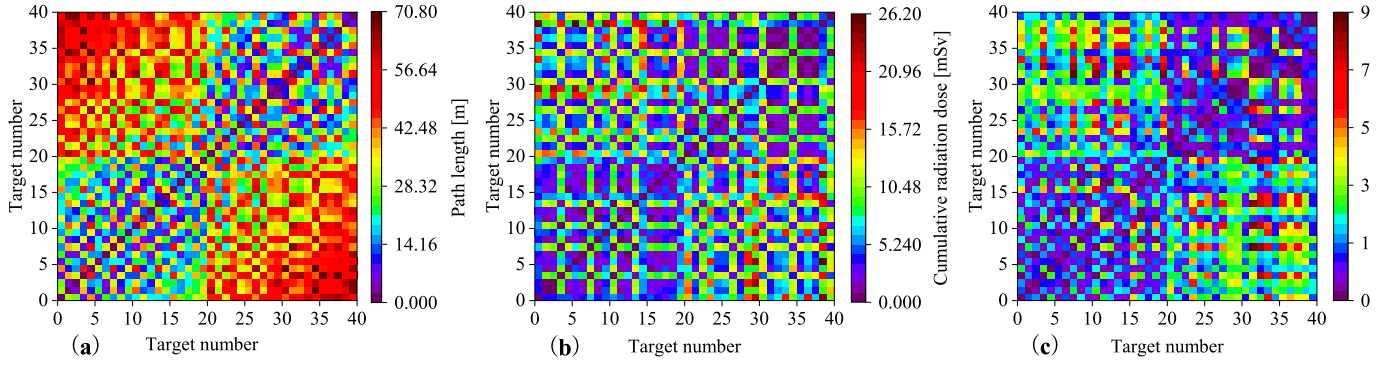


Fig. 15. Cost graphs of scenario 6 obtained by modified A*. (a) path length; (b) risk degree; (c) turning energy consumption.

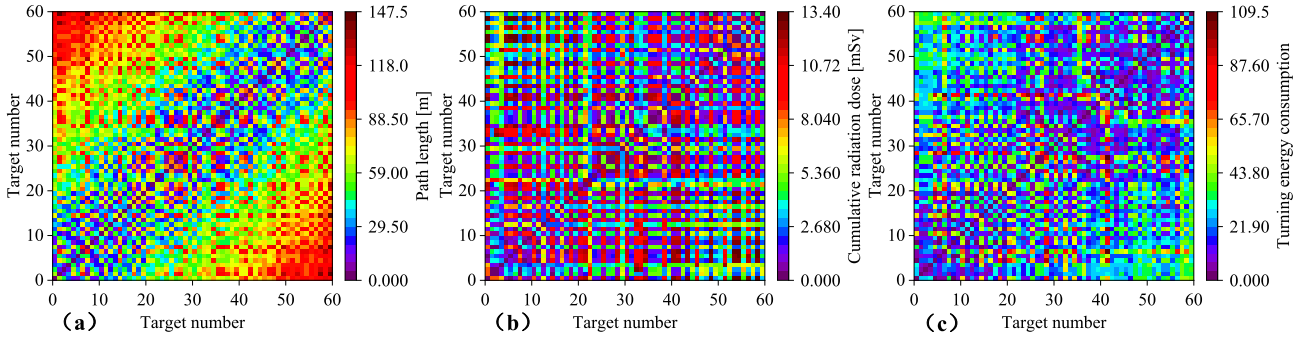


Fig. 16. Cost graphs of scenario 7 obtained by modified A*. (a) path length; (b) risk degree; (c) turning energy consumption.

Table 7

Parameters of the comparison algorithms in solving MTIPP problem.

Scenarios ID	Algorithm	Parameters
4	ACO-A*	$I_{max} = 200, N_{max}=100, \alpha = 1, \beta = 5, \rho = 0.3, \tau_0=0.5$
	GA-A*	$G_{max} = 200, n_p = 100, p_c=0.9, p_m=0.05, gap=0.9$
	GA-ACO-A*	$G_{max} = 200, n_p = 100, p_c=0.9, p_m=0.05, gap=0.9$
	ACO-GA-A*	$I_{max} = 200, N_{max}=100, \alpha = 1, \beta = 5, \rho = 0.1, \tau_0=0.5$
5-7	ACO-A*	$I_{max} = 500, N_{max}=200, \alpha = 1, \beta = 5, \rho = 0.1, \tau_0=0.5$
	GA-A*	$G_{max} = 500, n_p = 200, p_c = 0.9, p_m=0.05, gap=0.9$
	GA-ACO-A*	$G_{max} = 500, n_p = 200, p_c = 0.9, p_m=0.05, gap=0.9$
	ACO-GA-A*	$I_{max} = 500, N_{max}=200, \alpha = 1, \beta = 5, \rho = 0.1, \tau_0=0.5$

complexity, the shortest path, safest path and minimum turning energy consumption path can be obtained by the change of weights, which denotes that the optimal trade-off collision-free path traversing all targets will be generated by adjusting the weights value; 2) ACO-GA-A* outperforms comparison algorithm in finding the shortest path in each scenarios, with its solutions relatively concentrated. Concerning the safest path, ACO-GA-A* achieves superior results compared to other

algorithms, except for the solutions in scenario 4 are more dispersed than those of GA-ACO-A*. The optimal path with minimum energy consumption planned by ACO-GA-A* demonstrates superiority over other algorithms, except for scenario 7 where the solutions are more dispersed than those of GA-ACO-A*; 3) Considering all the three constraints, i.e., weights are set to (1, 1, 1), the optimal inspection path with minimum total cost obtained by our proposed algorithm is obviously superior to other algorithms in different scenarios; The shortest paths planned by the proposed algorithm in every scenarios are superior to the comparison algorithms, and have smaller standard deviations. The analysis above suggests that the changes in weight values and alterations in environmental complexity will not affect the effectiveness of the proposed algorithm.

On the other hand, the performance of ACO-GA-A* in solving MTIPP problems is affected by multiple control parameters, especially the pheromone concentration factor α , expected heuristic factor β and pheromone volatile factor ρ . To analyze the sensitivity of these control parameters to the total cost of planned path and the number of iterations, the scenario 4 is taken as an example and the cost coefficient weights is set as (1, 1, 1). The default parameters of the algorithm are set as follows: $\alpha = 3, \beta = 5, \rho = 0.7, Q = 1, \epsilon_1 = 0.01, \epsilon_2 = 0.09$. The control variable method is used for the simulation experiment, and each group

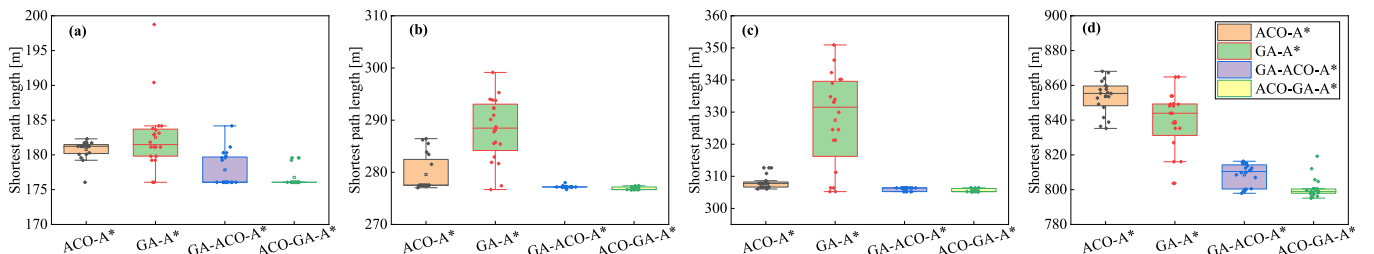


Fig. 17. The boxplots of the shortest paths obtained by different algorithms under different scenarios. (a) scenario 4; (b) scenario 5; (c) scenario 6; (d) scenario 7.

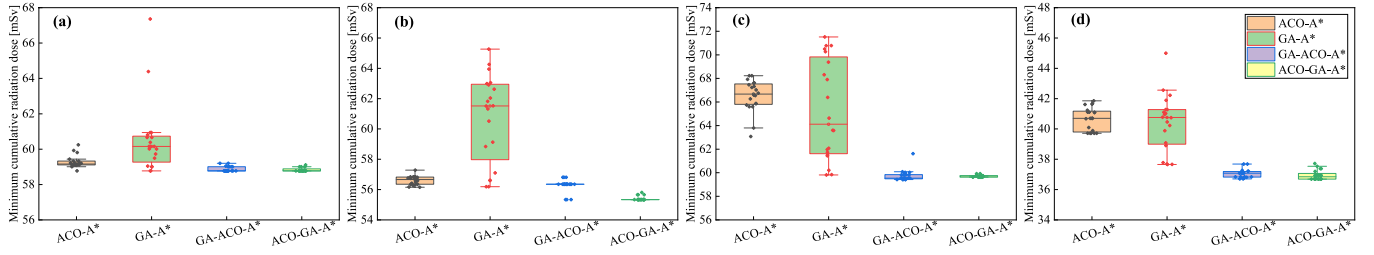


Fig. 18. The boxplots of the safest paths obtained by different algorithms under different scenarios. (a) scenario 4; (b) scenario 5; (c) scenario 6; (d) scenario 7.

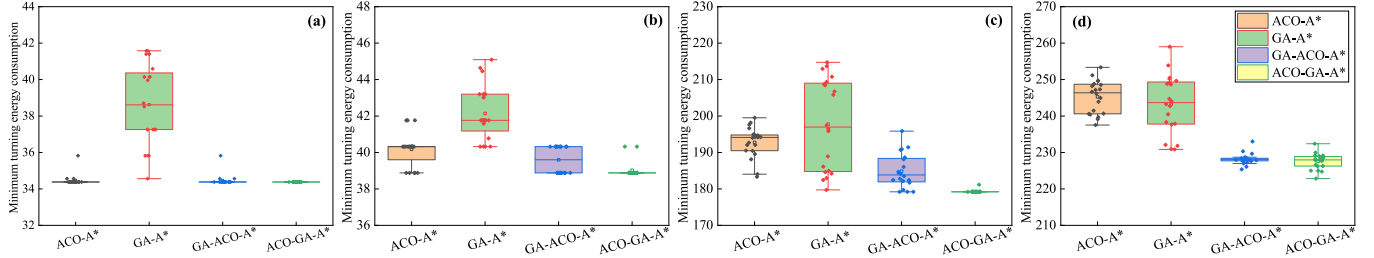


Fig. 19. The boxplots of the paths with minimum turning energy consumption obtained by different algorithms under different scenarios. (a) scenario 4; (b) scenario 5; (c) scenario 6; (d) scenario 7.

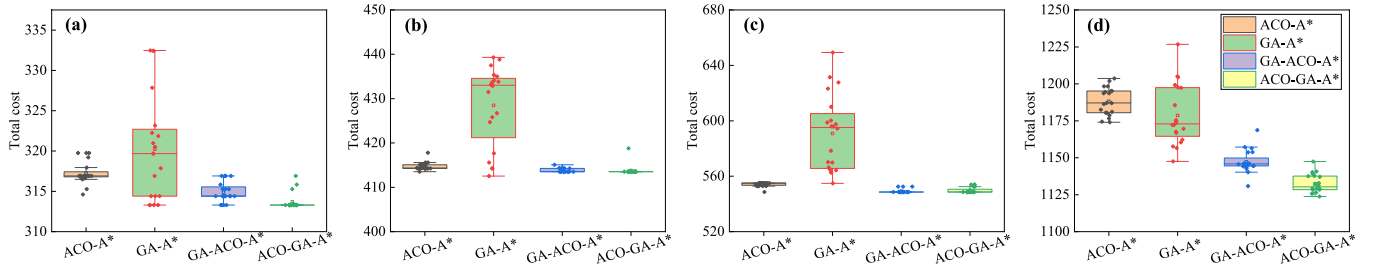


Fig. 20. The boxplots of the three-constraint paths obtained by different algorithms under different scenarios. (a) scenario 4; (b) scenario 5; (c) scenario 6; (d) scenario 7.

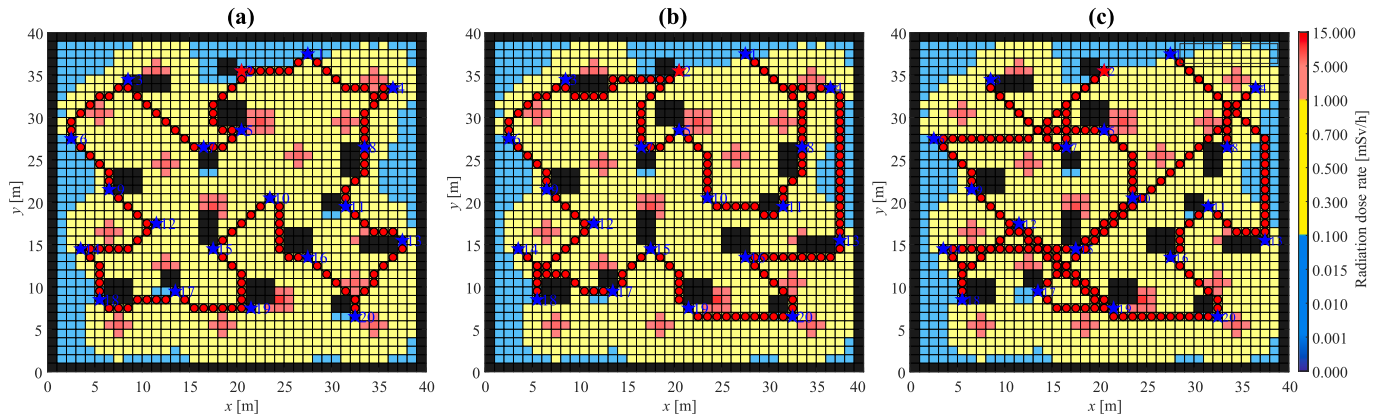


Fig. 21. Paths planned by ACO-GA-A* under different constraints in scenario 4. (a) the shortest path; (b) the safest path; (c) the path with minimum turning energy consumption.

of parameters is simulated 10 times. After simulation experiments, the error bar diagram describing the influence of control parameters on the total cost and number of iterations is shown in Fig. 25. As can be seen from Fig. 25, when β and ρ is constant, α has a little influence on the total cost, and the standard error of total cost and the number of iteration is smaller when α is smaller. When α , ρ is constant, the total cost decreases with the increase of β , and the standard error decreases, and the change

of β has little effect on the number of iterations. When α and β are constant, ρ has little effect on the total cost of the path and the number of iterations, and the standard error of the number of iterations decreases with the increase of ρ . It can be seen from the above analysis that β and ρ of the algorithm should be set larger, while α should be set smaller to improve the quality of the planned inspection path.

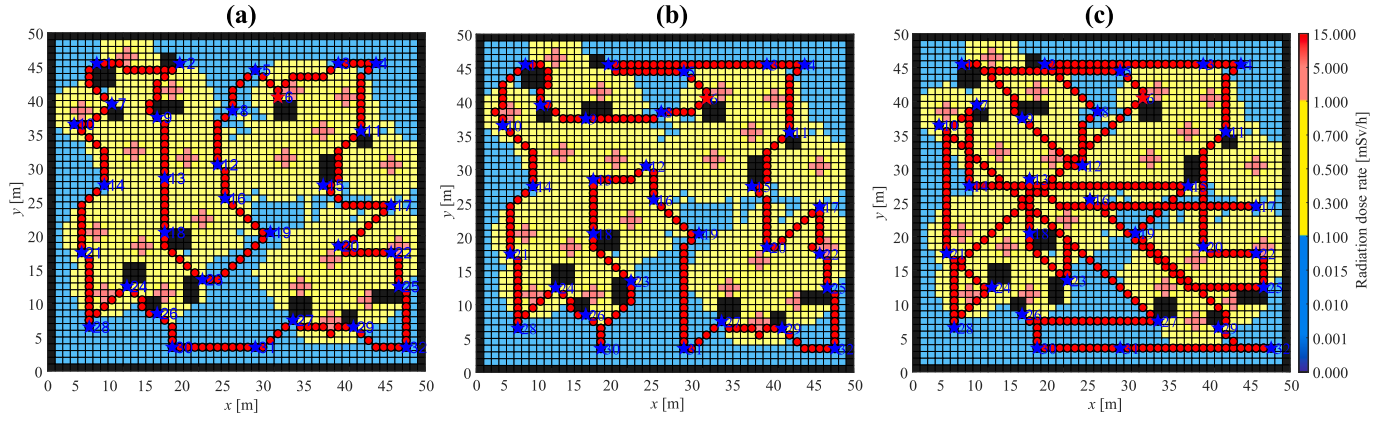


Fig. 22. Paths planned by ACO-GA-A* under different constraints in scenario 5. (a) the shortest path; (b) the safest path; (c) the path with minimum turning energy consumption.

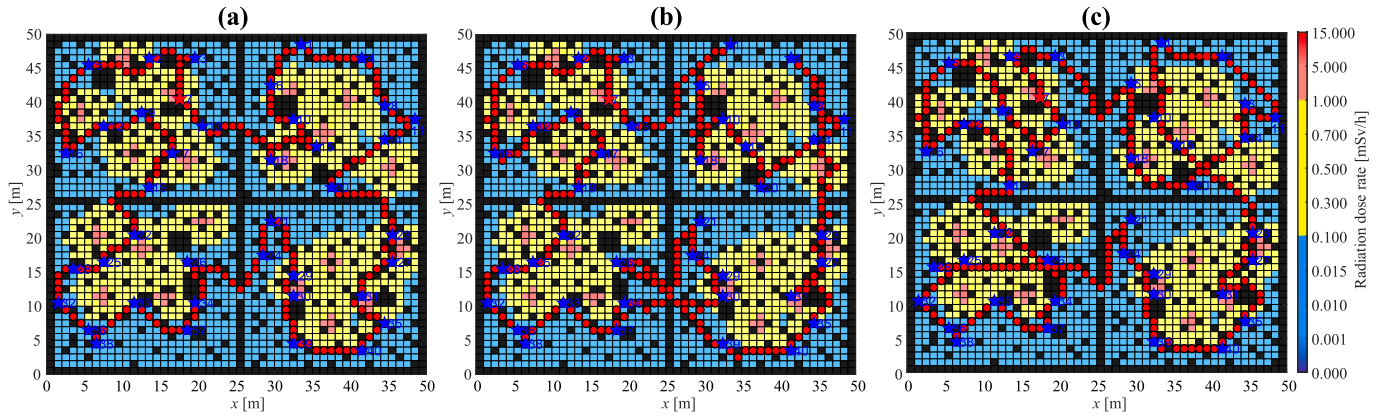


Fig. 23. Paths planned by ACO-GA-A* under different constraints in scenario 6. (a) the shortest path; (b) the safest path; (c) the path with minimum turning energy consumption.

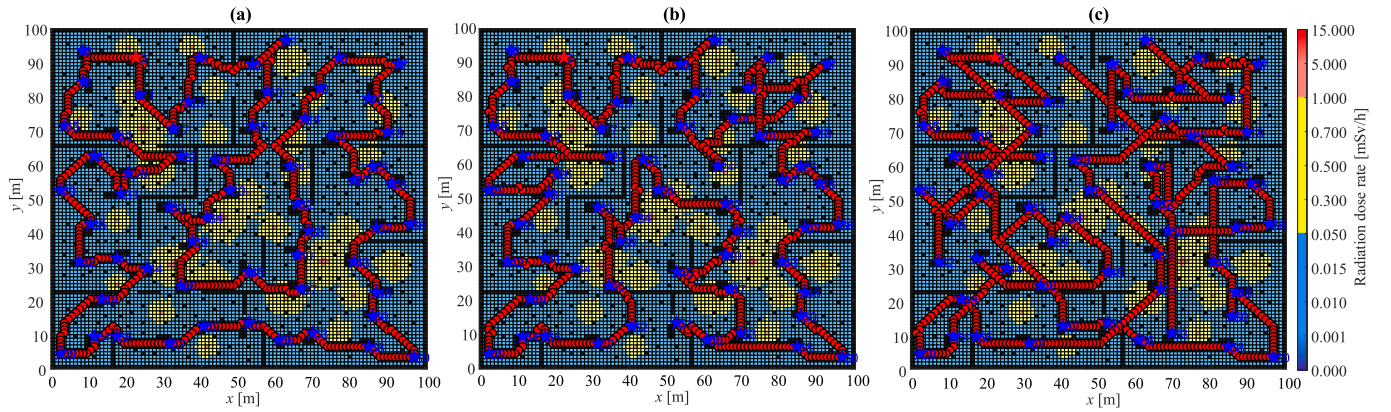


Fig. 24. Paths planned by ACO-GA-A* under different constraints in scenario 7. (a) the shortest path; (b) the safest path; (c) the path with minimum turning energy consumption.

6.4.2. Reliability analysis

To test the reliability of the proposed algorithms under different complex radioactive indoor environment, simulation experiments based on the near-natural hypothetical environments 4–7 described as section 6.1 are performed. As shown in Figs. 17–19, in different environments, path length (L), risk degree (R) and turning energy consumption (E) have different orders of magnitude. To balance the trade-off between the multiple objective constraints, the appropriate values of the cost weight coefficients ω_1 , ω_2 and ω_3 should be selected to standardize the three

single-objective functions (Miao et al., 2021). Firstly, the optimal value of L , R and E are calculated separately, as shown as Figs. 17–19. Then, the weight coefficients of the other two are determined separately in terms of the order of magnitude of the path length L , e.g., $\omega_1 = 1$, $\omega_2 = L_{min}/R_{min}$, $\omega_3 = L_{min}/E_{min}$. The reliability analysis based on five indicators of the average total cost, maximum total cost, minimum total, average computational time and proportion of optimal value are recorded in Table 8. The optimal multi-target inspection path, traversal order and total cost planned by these algorithms in different scenarios

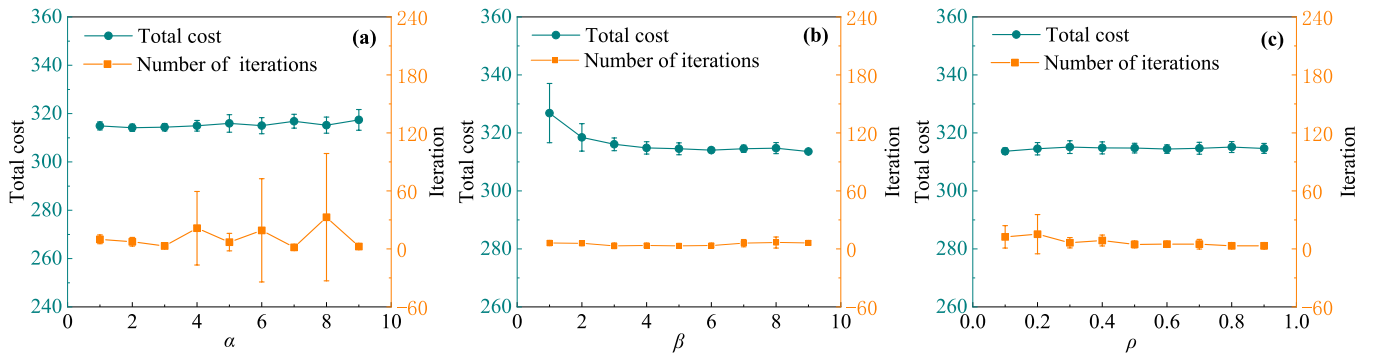


Fig. 25. The sensitivity of control parameters of ACO-GA-A* on total cost and iteration number when solving MTIPP problem. (a) α [$\beta = 5, \rho = 0.7$]; (b) β [$\alpha = 3, \rho = 0.7$]; (c) ρ [$\alpha = 3, \beta = 5$].

Table 8

Reliability analysis results of the proposed algorithms under different scenarios. The values of average total cost and average computational time are reported as the mean value $\pm$ standard deviation for 20 executions.

Scenario No.	Cost weight coefficients	Algorithms	Average total cost	Maximum total cost	Minimum total cost	Average computational time (s)	Proportion of optimal value (%)
4	$\omega_1 = 1.0$ $\omega_2 = 3.0$ $\omega_3 = 3.6$	ACO-A*	606.778 $\pm$ 7.24	616.375	598.877	23.47 $\pm$ 0.43	35
		GA-A*	610.253 $\pm$ 9.85	629.123	598.575	21.22 $\pm$ 0.52	25
		GA-ACO-A*	604.956 $\pm$ 5.72	617.028	598.575	24.50 $\pm$ 0.97	30
		ACO-GA-A*	598.575 $\pm$ 0.00	598.575	598.575	24.27 $\pm$ 0.48	100
5	$\omega_1 = 1.0$ $\omega_2 = 4.7$ $\omega_3 = 4.1$	ACO-A*	912.470 $\pm$ 2.15	921.586	911.990	111.14 $\pm$ 1.01	95
		GA-A*	943.775 $\pm$ 21.56	9078.402	900.406	87.08 $\pm$ 1.75	5
		GA-ACO-A*	902.5134 $\pm$ 4.32	910.943	900.406	120.05 $\pm$ 0.63	80
		ACO-GA-A*	900.406 $\pm$ 0.00	900.406	900.406	119.67 $\pm$ 1.53	100
6	$\omega_1 = 1.0$ $\omega_2 = 4.0$ $\omega_3 = 2.0$	ACO-A*	960.707 $\pm$ 10.20	977.717	935.963	149.64 $\pm$ 0.87	5
		GA-A*	941.757 $\pm$ 33.25	975.711	898.238	113.82 $\pm$ 0.59	10
		GA-ACO-A*	904.855 $\pm$ 5.40	921.009	898.238	156.03 $\pm$ 0.63	15
		ACO-GA-A*	901.896 $\pm$ 4.21	905.297	897.036	156.74 $\pm$ 0.64	30
7	$\omega_1 = 1.0$ $\omega_2 = 23.7$ $\omega_3 = 2.7$	ACO-A*	2794.140 $\pm$ 31.90	2838.410	2759.320	1978.69 $\pm$ 118.22	5
		GA-A*	2775.821 $\pm$ 81.55	2919.920	2678.080	1862.19 $\pm$ 34.39	5
		GA-ACO-A*	2628.705 $\pm$ 23.27	2696.660	2611.560	1898.73 $\pm$ 92.97	10
		ACO-GA-A*	2613.925 $\pm$ 8.38	2628.490	2603.290	1894.78 $\pm$ 139.41	20

are shown in Figs. 26–29.

It can be seen from Table 8 that all of the average total cost and its standard deviation of paths planned by the proposed bi-layer hybrid algorithm are significantly superior to other algorithms in different complexity of scenarios. Moreover, the probability of obtaining the optimal path under the environments 4–6 reaches 100 %, 100 %, 30 % and 20 %, which is significantly higher than other algorithms. The results shows that the proposed algorithm performs better in terms of reliability and robustness compared with other algorithms. In terms of computational time, the hybrid algorithms (i.e., ACO-GA-A* and GA-ACO-A*) have no advantages over the ACO-A* and GA-A*. The reason is that it will take more time cost to enhance the initial pheromone distribution of ACO by GA or to improve the initial population of GA by ACO. For the MTIPP problem, which is an offline PP carried out before MR enters the inspection site, it is acceptable to obtain a superior solution through a small amount of time cost.

6.4.3. Convergence characteristic analysis

Fig. 30 depicts the convergence curves of different algorithms for planning the optimal multi-target inspection paths under different radioactive scenarios. It can be seen that the proposed algorithm converges to the optimal solution faster, while other algorithms' convergence speed is slower and cannot find the optimal inspection path with minimum total cost.

The convergence curves of traditional intelligent algorithms (i.e., ACO and GA) with A* changes with the increase of the complexity of the environment, as shown in Fig. 16. We can see that the performance of ACO-A* decreases rapidly with the complexity of the environment increases, especially in the optimal solution. Although, the GA-A* algorithm converge faster than ACO-A* in the early stage, its convergence speed slows down in the late stage and does not find the optimal path finally. The number of iterations required for GA-A* convergence is significantly increased in the complex environments.

Comparing the convergence curves of ACO-GA-A* and ACO-A*, it

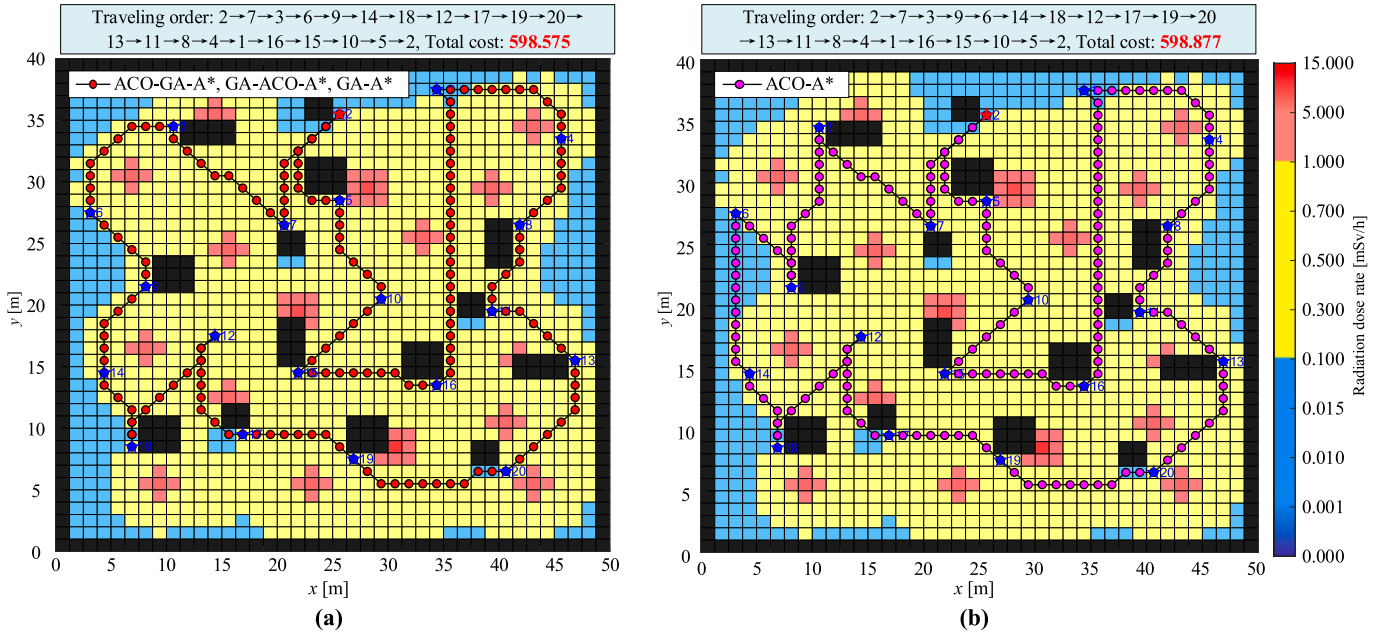


Fig. 26. The optimal multi-target inspection path planned by different algorithms in scenario 4. (a) ACO-GA-A*, GA-ACO-A* and GA-A*; (b) ACO-A*.

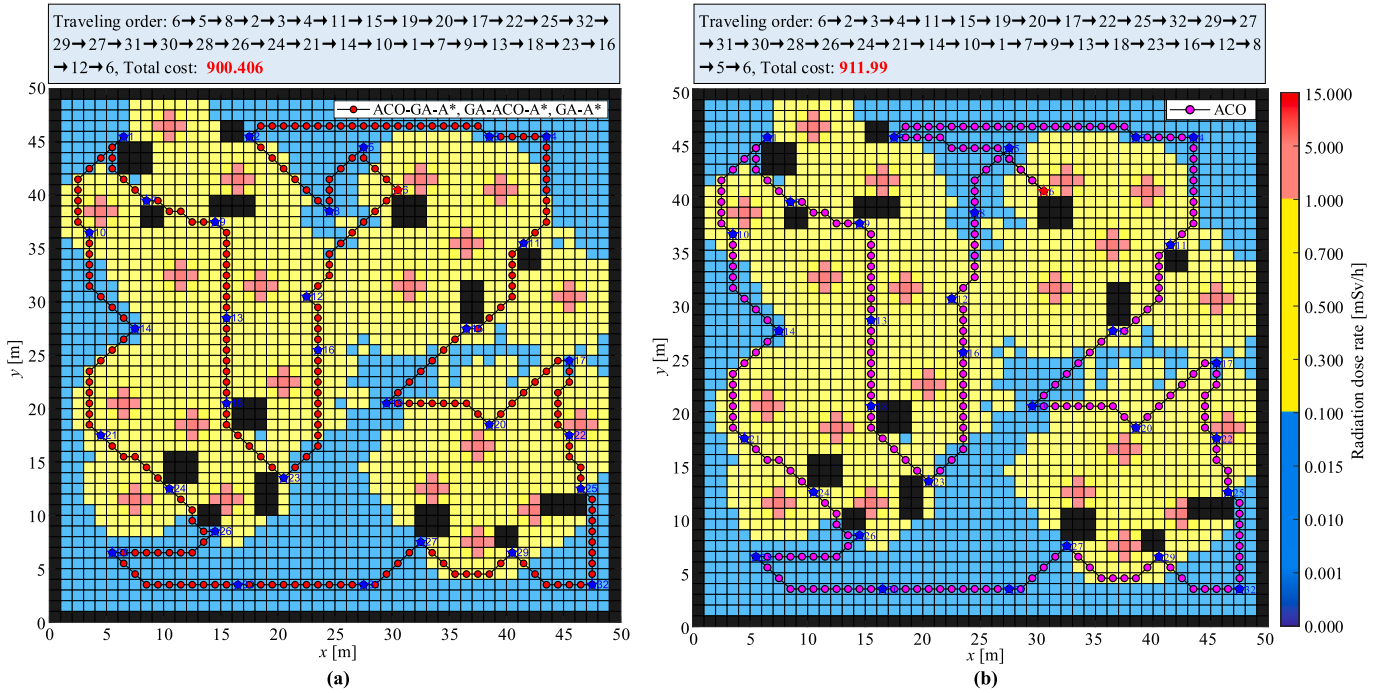


Fig. 27. The optimal multi-target inspection path obtained by different algorithms in scenario 5. (a) ACO-GA-A*, GA-ACO-A* and GA-A*; (b) ACO-A*.

can be confirmed that the enhancement of ACO' initial pheromone guidance by GA mentioned in Section 5.1.1 has a significant effect in improving the convergence rate. Meanwhile, comparing the convergence of GA-ACO-A* and GA-A*, we can see that the convergence rate of traditional GA is improved significantly by the optimization of initial population of GA based on ACO algorithm. In addition, it is no doubt that the proposed hybrid algorithm is easier to find the optimal solution compare with GA-ACO-A*.

6.5. Experiment in real-world environment

To evaluate the efficacy of the proposed approach, it is applied to the task of inspection PP for a MR within an experimental setup designed to mimic a NPP environments, as illustrated in Fig. 31. The inspection MR is equipped with an RS-Helios-16P LiDAR sensor and a radiation detector sensor to obtain external environmental data prior to the commencement of PP, as shown in Fig. 1 (a). Within this environment,

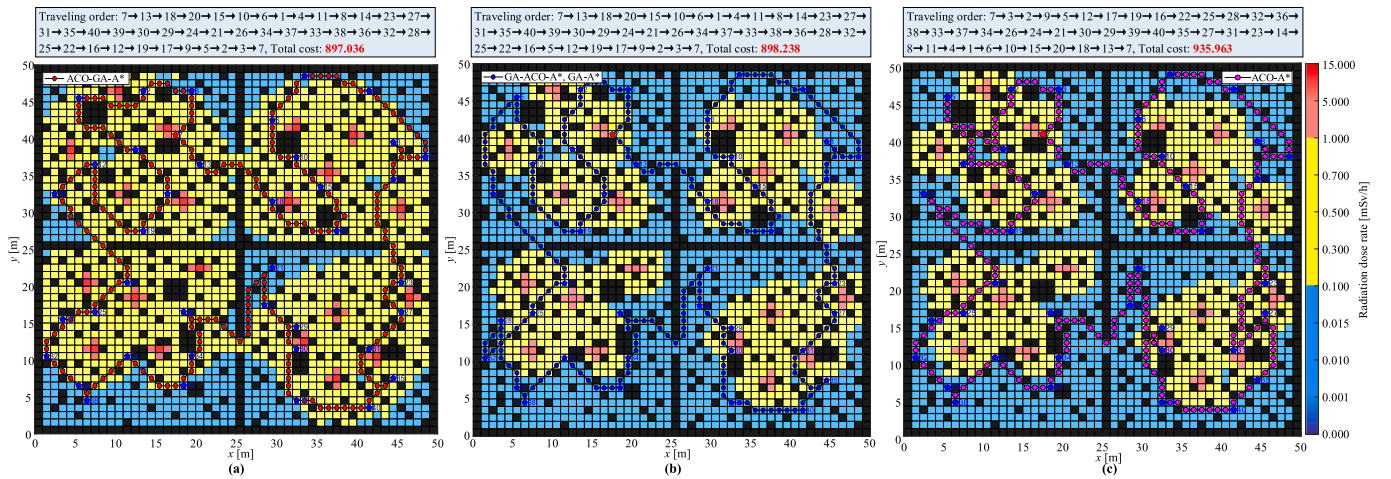


Fig. 28. The optimal multi-target inspection path planned by different algorithms in scenario 6. (a) ACO-GA-A*; (b) GA-ACO-A* and GA-A*; (c) ACO-A*.

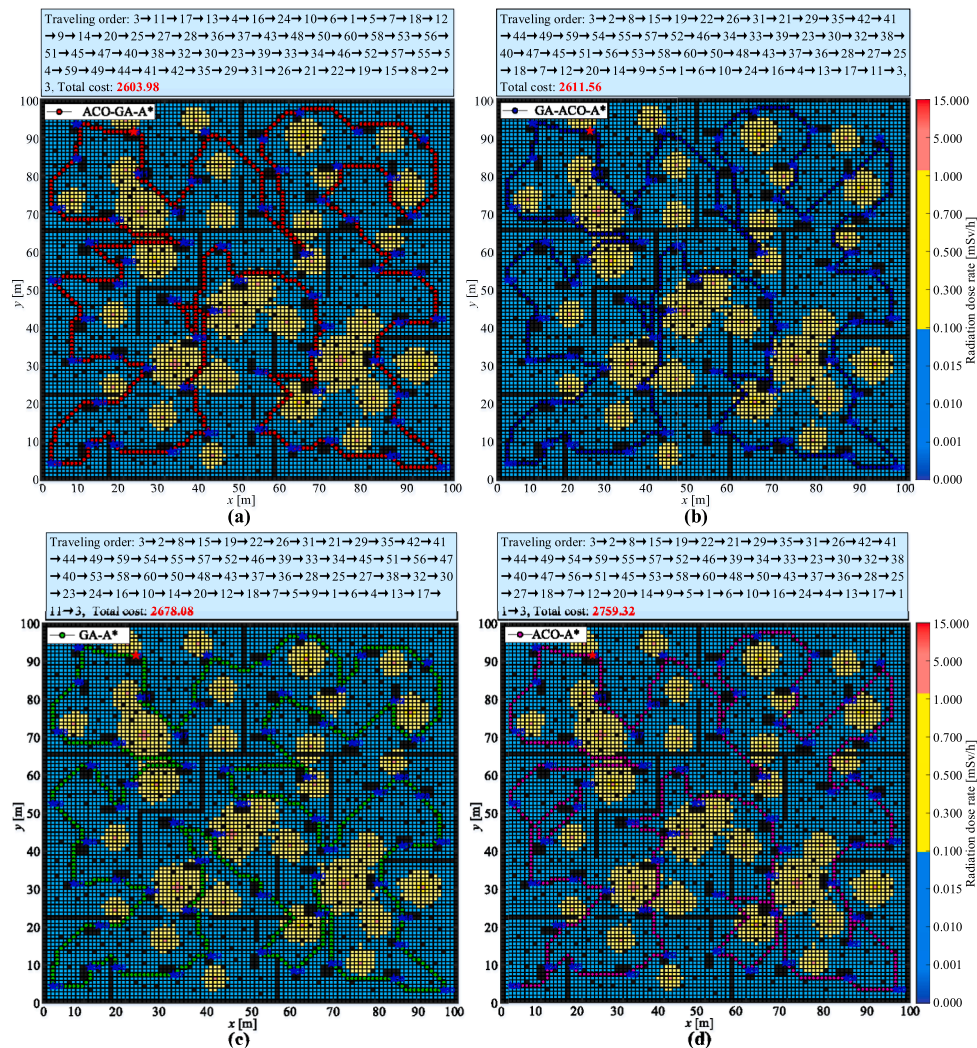


Fig. 29. The optimal multi-target inspection path planned by different algorithms in scenario 7. (a) ACO-GA-A*; (b) GA-ACO-A*; (c) GA-A*; (d) ACO-A*.

two identical γ -ray sources are positioned, each characterized by an energy output of 1.25 MeV and an activity level of 1.85×10^5 Bq. Additionally, the environment incorporates two impassable steel cylinders, representing nuclear facility components with a shielding effect, and a single passable obstacle step. For the purposes of analysis, the

environment's lower-left corner was designated as the origin of coordinates, with the camera's perspective fixed at the upper-right corner for the duration of the experiment. Six inspection targets, labeled T_1 (1.25, 7.75), T_2 (6.75, 7.75), T_3 (3.75, 3.75), T_4 (1.25, 1.25), T_5 (4.75, 1.25) and T_6 (8.25, 1.25), are established within this

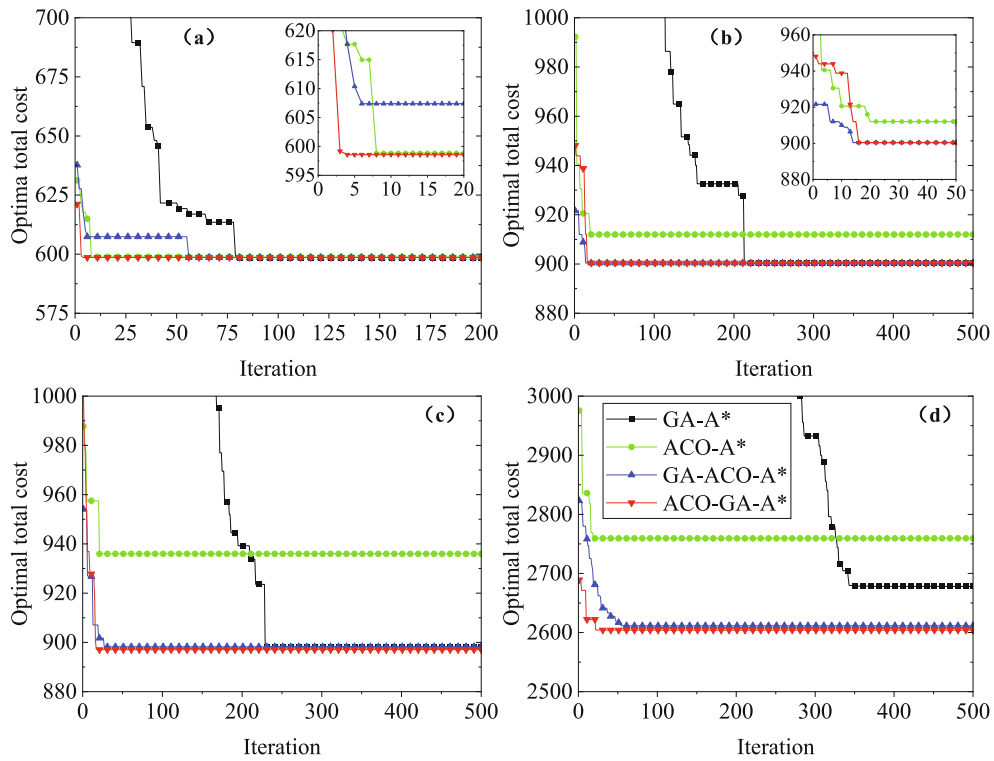


Fig. 30. Coverage curves of different algorithms for planning multi-target optimal inspection paths under different scenarios. (a) scenario 4; (b) scenario 5; (c) scenario 6; (d) scenario 7.

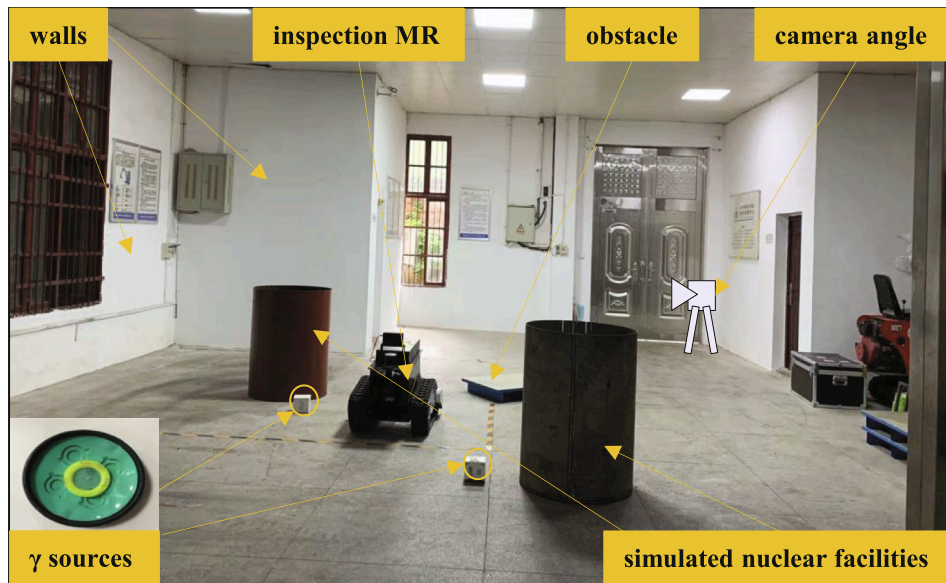


Fig. 31. The experimental environment simulating the real NPP environments.

realistically simulated setting, with T_6 serving as the initial inspection point.

The inspection path planned by the proposed algorithm ACO-GA-A* is shown in Fig. 32. As can be seen from the Fig. 32, the traveling order of the inspection targets is ($T_6 \rightarrow T_2 \rightarrow T_1 \rightarrow T_4 \rightarrow T_3 \rightarrow T_5 \rightarrow T_6$). Furthermore, the same inspection targets traversal order is set at the robot operating

system (ROS) end of the inspection MR for PP, and the planning process is shown in Fig. 33. The comparison of simulation and experimental results in solving MTIPP problem is recorded in Table 9. As can be seen from Table 9, the real experimental data is slightly larger than the simulation value, mainly because there are errors in the path length calculated by the odometer of the inspection MR, and when the MR

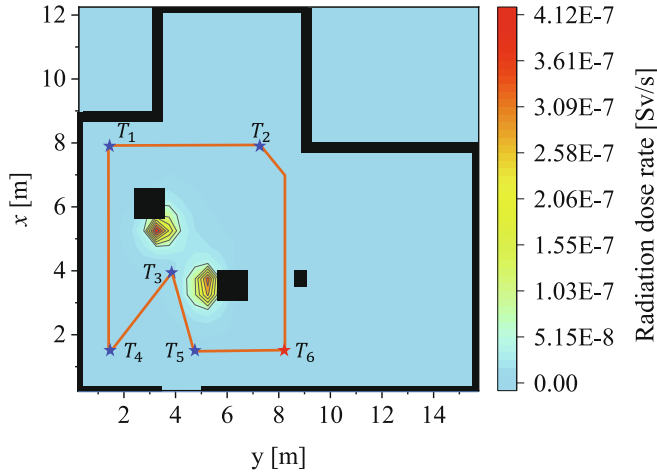


Fig. 32. Inspection path planned by ACO-GA-A* in simulation experiment.

reaches an inspection target, the pose adjustment is required to guide the MR to the next inspection target or the destination target, which will consume more time and absorb more radiation doses. However, it can be seen from Fig. 33 that the real inspection PP effect is close to the simulation effect, and the multi-target inspection task can be performed according to the predetermined traversal order.

7. Conclusion

In this paper, to address the MTIPP problems of MRs in indoor environments with multiple radioactive sources and dense obstacles, a bi-level programming framework is constructed. The upper-level is a MOTSP traversing all inspection targets, while the low-level is a pairwise MOPP between each inspection target. To solve the proposed constructed model, a bi-level planner called ACO-GA-A* is proposed by

Table 9

Comparison of simulation and experimental results in solving MTIPP problem.

	Path length (m)	Cumulative radiation dose (Sv)	Time (s)
Simulation	29.07	1.66×10^{-7}	58.14
Real experiment	32.43	1.84×10^{-7}	75.68

hybridization ACO-GA incorporated with the deterministic search algorithm modified A*. In ACO-GA-A*, GA-ACO is responsible for optimizing the traveling order of inspection targets based on a two-layer grid cost map containing information about radioactive sources and obstacles. Following the traveling order obtained by ACO-GA, modified A* performs pairwise MOPP for providing cost graphs. And, the information interaction of each layer continuously until the optimal solution converges.

The effectiveness of the ACO-GA in the upper-level programming for optimizing the traveling order of multiple inspection targets has been validated by the simulation experiments based on TSPLIB examples. The superiority of the modified A* in the lower-level programming has been verified by comparing with original A*, modified ACO and GA in three scenarios. In addition, the sensitivity, reliability and convergence characteristic of ACO-GA-A* have been demonstrated through comparative experiments with ACO-A*, GA-A* and GA-ACO-A* in different complexity radioactive indoor environments. Finally, the ACO-GA-A* algorithm has been applied to the inspection MR equipped with LiDAR and radiation detector, and the experimental results demonstrate that the bi-level programming framework for MTIPP and the proposed algorithm is both effective and feasible.

The focus of future work is to optimize the parameter combination of the proposed algorithm with the help of intelligent algorithms to improve solution quality of MTIPP. Another research direction is how to extend the proposed algorithm to 3D radioactive environment or dynamic radioactive environment, and apply it to other fields, such as radioactive materials road transportation (Tao et al., 2022) and radioactive sources detection (Brouwer et al., 2021).

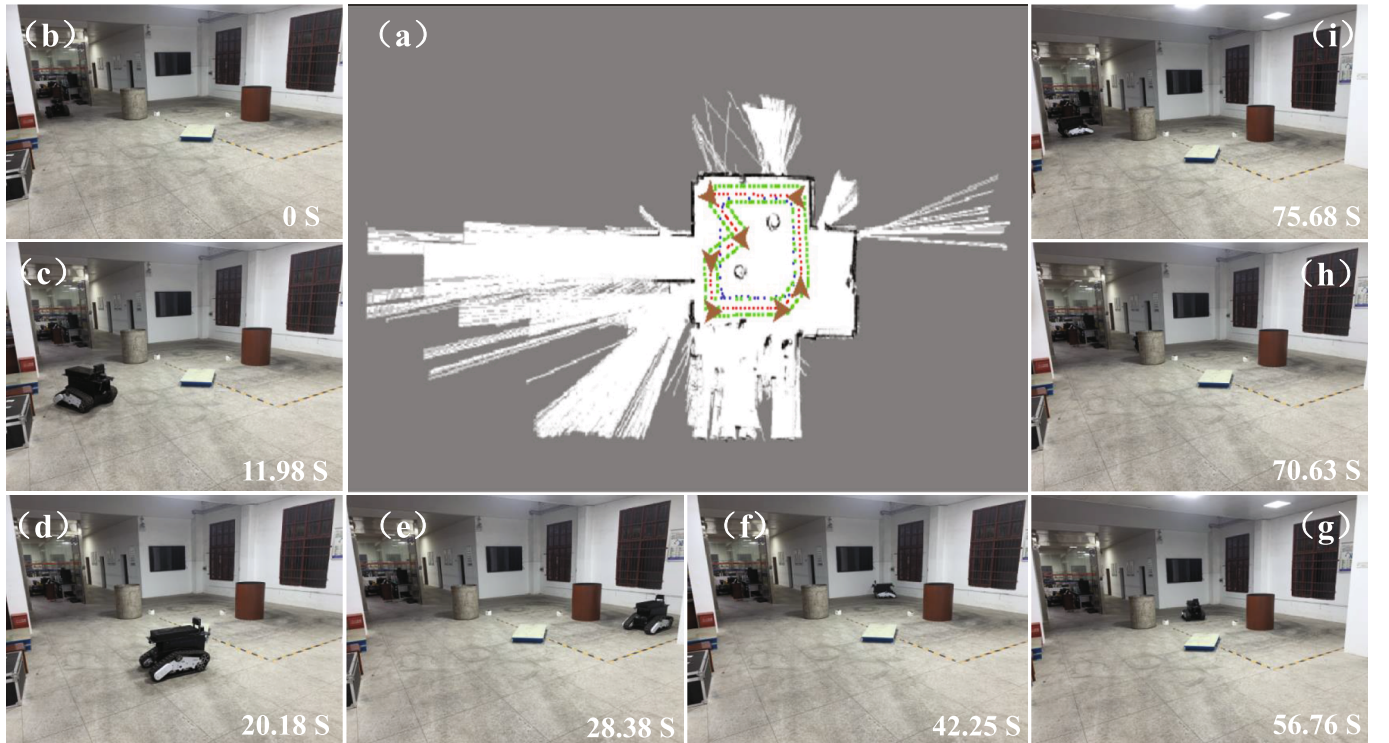


Fig. 33. PP process of the inspection MR in the experimental environment. (a) inspection path; (b) robot located at T_6 ; (c) arrive at the turning point between T_6 and T_2 ; (d) arrive at T_2 ; (e) arrive at T_1 ; (f) arrive at T_4 ; (g) arrive at T_3 ; (h) arrive at T_5 ; (i) return to T_6 .

CRediT authorship contribution statement

De Zhang: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Validation, Resources, Software, Data curation, Project administration, Funding acquisition. **Yebo Yin:** Software. **Guokai Xiong:** Software. **Shuliang Zou:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eswa.2024.126095>.

Data availability

Data will be made available on request.

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